Integration of Artificial Intelligence in Secondary School Science Teaching: A Systematic Review of the Research 2000 – 2024

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Abstract

This systematic review investigates the integration of Artificial Intelligence in secondary school science teaching from 2000 to 2024, focusing on its impact on pedagogical practices and student learning outcomes. Through a mixed-method approach, the study synthesizes findings from 10 studies to identify key AI applications, such as intelligent tutoring systems, adaptive learning platforms, and virtual laboratories. The results demonstrate that AI enhances personalized learning, improves student engagement, and supports deeper conceptual understanding. However, challenges like teacher preparedness, infrastructure limitations, and data privacy concerns persist. The review provides recommendations for educators, policymakers, and researchers to ensure effective AI adoption in education, emphasizing the need for professional development and equitable resource distribution. The study concludes by proposing future research directions to address the identified challenges.

Keywords:

Artificial Intelligence, secondary school science, science teaching, AI applications, educational technology, teacher preparedness

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Abbreviations

AI – Artificial Intelligence

AIE - Artificial Intelligence in Education

AISE - Artificial Intelligence in Science Education

CASP – Critical Appraisal Skills Programme

CAT - Computer-aided translation

ICT – Information and Communication Technology

ITS – Intelligent Tutoring Systems

PRISMA – Preferred Reporting Items for Systematic Reviews and Meta-Analyses

STEM – Science, Technology, Engineering, and Mathematics

WoS – Web of Science

TAM - Technology Acceptance Model

Chapter One
Introduction to the Study

1.1 Introduction – Background and Rationale

In today's rapidly changing world, innovation plays crucial role in shaping teaching and learning practices, especially the advancement of technology. In contrast, earlier education was shaped without the technological advancements we have today (Chen et al., 2020). In the past, classrooms relied heavily on traditional teaching methods, such as lectures, textbooks, and handwritten assignments (Balliu, 2017). Furthermore, educators spent countless hours manually grading papers, preparing lesson plans, and delivering lectures with limited interactive tools. This traditional approach often results in one-size-fits-all instruction, with little room for personalized learning experiences or real-time feedback (Balliu, 2017). As we look back on these practices, it becomes evident that the absence of technological advancements hindered the efficiency and effectiveness of secondary school science teaching (Chen et al., 2020).

Looking at the evolution of educational practices, we observe a significant departure from the traditional methods of teaching commonly used in earlier decades (Jia et al., 2020). Roll and Wylie (2016) elucidate, as technology began to advance and permeate educational settings, educators gained access to an abundance of digital tools and resources that revolutionized their approach to instruction (Roll and Wylie, 2016). Moreover, the revolution is a trajectory shaped by incremental advancements and transformative innovations. Initially rooted in the exploration of programmed instructions, the evolution of computer technology catalysed its integration into educational settings, ushering in an era of computer-aided instruction and learning (Eaton et al., 2018). As computing capabilities matured, the advent of networking, the internet, and sophisticated software solutions broadened the horizons of educational technology, enabling dynamic interactions and access to vast repositories of knowledge (Dobrev, 2012). Moreover, Roll and Wylie (2016) write, the culmination of these developments heralded the dawn of artificial intelligence, a ground-breaking paradigm shift characterized by machines exhibiting human-like cognitive abilities. This narrative underscores the dynamic interplay between technological innovation and pedagogical practices, reshaping the landscape of education.

Artificial intelligence has been harnessed in secondary school science teaching, leveraging elements like machine learning and intelligent agent technology to enhance various aspects of the learning process (Perrakis and Sixma, 2021; Cheah, 2021). Its adoption has led to notable improvements in efficiency, personalized learning experiences, content delivery, and administrative processes within educational institutions (Jia et al., 2020). Within secondary school science teaching, Artificial Intelligence presents an array of promising prospects aimed at elevating pedagogical practices and enriching learning encounters.

Rationale for the Research

In many regions, the education system faces significant challenges, particularly in the effective teaching and learning of science subjects in secondary schools. A lack of quality teaching materials, unequal access to educational resources, and pervasive inequities have hindered the ability of many students to fully engage with and succeed in science education. These challenges are exacerbated by the lack of well-equipped science laboratories, insufficient teacher training, and inadequate curriculum resources, which create disparities in learning outcomes between students from different socio-economic backgrounds (Krajcik, 2002). These disparities have led to significant gaps in student performance, with many learners unable to access the same level of instruction and support as their peers in better-resourced schools (Bang & Brown, 2022; Bell & Bang, 2022). AI holds the potential to bridge these gaps by offering innovative, scalable, and equitable solutions to the challenges facing science education in the country. The flexibility and accessibility of AI tools available anytime to anyone with the necessary digital infrastructure can help democratize access to quality learning resources (Parveen & Ramzan, 2024).

This research is motivated by the belief that the integration of AI into secondary school science teaching can revolutionize the way science subjects are taught and learned. By empowering educators to harness the full potential of AI technologies, we can enhance the learning experience, making it more personalized, interactive, and engaging. Moreover, AI can provide learners with continuous access to information and learning tools, irrespective of their physical location or the limitations of their schools' resources. This research seeks to explore how AI can be strategically integrated into science teaching to foster greater learner interest in science subjects, improve educational outcomes, and reduce the resource disparities that have long characterized the South African educational landscape. With appropriate training and support,

educators can become catalysts for change, using AI to unlock new opportunities for their students.

1.2 Relevance and Significance of the Research

In the secondary school science teaching sector, artificial intelligence, as identified by Liu et al (2021), has found applications in administration, teaching, and learning. Teaching and learning, as the framework for comprehending artificial intelligence's role in secondary school science teaching, will constitute the focus of this study.

The integration of AI in education is a crucial advancement in addressing persistent challenges in the teaching and learning of science subjects in secondary schools. This study is particularly relevant given the widespread disparities in access to quality education, especially in under-resourced environments. The lack of adequate teaching materials, well-equipped laboratories, and trained educators contributes to significant inequities in learning outcomes among students from diverse backgrounds. By focusing on the potential of AI to bridge these gaps, this research provides valuable insights into innovative, scalable, and equitable educational solutions.

Existing literature has highlighted significant gaps in understanding how AI can effectively enhance science education, particularly in settings where resources are limited. This study aims to fill these gaps by examining the effectiveness of AI applications in promoting inclusive and engaging science teaching practices. By exploring the intersection of AI technology and pedagogical approaches, the research seeks to inform educators, policymakers, and stakeholders about the transformative potential of AI in improving educational equity.

The findings of this study hold practical significance for educators looking to leverage technology to foster student engagement and enhance learning experiences. As AI tools become increasingly accessible, understanding their role in democratizing science education is essential. This research not only aims to contribute to theoretical knowledge in the field of educational technology but also seeks to provide actionable recommendations for practitioners striving to create more inclusive and effective learning environments.

In an era marked by rapid technological advancements and an urgent need for equitable education, this study is timely and critical. The outcomes of this research could inform future educational policies and practices, ensuring that all students have the opportunity to succeed in science education, regardless of their background.

Significance of the Research

This research contributes to the growing body of literature on the integration of AI in education, specifically within the context of secondary school science teaching. While AI's potential in enhancing learning outcomes has been recognized globally, there is limited research focusing on its application in science education at the secondary school level, particularly in the South African context. This study will help fill this gap by providing a comprehensive review of how AI can be effectively utilized to address challenges such as limited access to resources, equity issues, and the overall effectiveness of teaching methodologies. By analyzing existing studies and synthesizing findings across two decades, this research aims to provide a nuanced understanding of AI's role in transforming science education, thereby contributing theoretically to ongoing discussions in both AI in education and science pedagogy.

From a practical standpoint, the findings of this study will have significant implications for teachers, students, policymakers, and curriculum developers. For educators, it will offer insights into how AI tools such as intelligent tutoring systems, adaptive learning platforms, and virtual labs can be integrated into the classroom to create more personalized, engaging, and effective learning experiences. Teachers will benefit from understanding how AI can supplement traditional teaching methods, allowing them to better support individual student needs. Policymakers and curriculum developers will gain evidence-based recommendations for the inclusion of AI in science curricula, paving the way for more informed decisions about the allocation of resources and the training of teachers. This research will also provide practical guidelines for implementing AI technologies in schools, ensuring that its use is equitable and enhances student learning outcomes across diverse contexts.

In addition to its immediate practical applications, this research has the potential to inspire further studies on the use of AI in secondary school education. Given that the integration of AI in education is still an emerging field, particularly in under-resourced regions, this study can serve as a foundation for future explorations into the long-term impacts of AI on student engagement, knowledge retention, and achievement in science subjects. It will encourage researchers to examine how AI can be applied across different educational settings and how it can be used to address specific challenges, such as teacher shortages, lack of materials, and disparities in educational quality. By laying the groundwork for future investigations, this research aims to contribute to the ongoing evolution of AI as a transformative tool

1.3 Research Problem

As education undergoes a technological transformation, the integration of Artificial Intelligence in secondary school science teaching has gained attention as an innovative approach to teaching and learning. While Artificial Intelligence applications are increasingly used in educational contexts, there is a need for a comprehensive understanding of how these technologies are implemented in secondary school science classrooms. Therefore, identifying gaps in the existing body of research is crucial for guiding future studies and enhancing pedagogical practices in secondary school science teaching.

Current literature often focuses on Artificial Intelligence's potential benefits, but there is limited empirical evidence on its effectiveness in improving student learning outcomes and engagement. Additionally, existing studies tend to concentrate on individual Artificial Intelligence applications, such as intelligent tutoring systems or adaptive learning platforms, without a broader analysis of their collective impact on science education. This lack of comprehensive studies creates a gap in our understanding of how AI can be effectively integrated into secondary school science classrooms, pointing to a need for a systematic review to gather and synthesize the available evidence.

Therefore, given the growing interest in AI's role in education, a systematic review of its integration into secondary school science teaching is crucial. While individual studies suggest that Artificial Intelligence can improve personalized learning, real-time feedback, and content delivery, there is a need for a comprehensive analysis to confirm these benefits empirically.

1.4 Research Objectives and Research Questions

This systematic review aims to synthesize and scrutinize existing research concerning the integration of AI in secondary school science teaching from 2000 to 2024. Through a comprehensive exploration of the literature landscape, this study seeks to identify diverse methodologies and Artificial Intelligence applications used, pinpoint areas of deficiency in existing research, and provide valuable insights to guide future scholarly investigations in this domain. Furthermore, as Artificial Intelligence continues to evolve, its applications in secondary school science teaching are likely to further revolutionize teaching and learning paradigms.

To systematically explore this topic, the following research questions have been formulated to guide this research.

Research Question 1: What research has been conducted on the integration of Artificial Intelligence in secondary school science education between 2000 and 2024?

Research Question 2: What Artificial Intelligence applications are commonly used in secondary school science education to enhance teaching and learning?

Research Question 3: How effective is the integration of Artificial Intelligence in secondary school science teaching?

Additionally, this systematic review seeks to address critical issues related to AI in secondary school science teaching, such as the range of Artificial Intelligence technologies in use, their effectiveness in improving learning outcomes. By synthesizing the current research landscape, this study aims to inform educators, researchers, and policymakers about the effective use of Artificial Intelligence in secondary school science teaching, guiding the future direction of this evolving field.

1.5 Overview of Conceptual Framework and Literature Review

This study is anchored in key concepts related to AI and its potential to transform secondary school science education. AI, for the purposes of this research, is understood as the simulation of human intelligence by machines, which includes processes such as learning, reasoning, and problem-solving. The scope of AI in education is broad, covering machine learning, natural language processing, virtual learning environments, and AI-driven tutoring systems. These technologies are increasingly being integrated into educational contexts to enhance both teaching and learning.

The secondary school science subjects that form the focus of this study include Life Sciences (Biology)/Natural Science/Human Biology, Combined Science, Physical Sciences, Chemistry, and Integrated Science. Each of these disciplines presents unique challenges and opportunities for AI integration. A critical aspect of this research is examining teacher preparedness to effectively incorporate AI tools into their teaching practices, as well as understanding the various AI applications being used in classrooms, such as intelligent tutoring systems and virtual laboratories. The effectiveness of these AI applications in improving student engagement, understanding, and academic performance is a key consideration, as this study aims to assess not just the implementation of AI but its tangible impact on learning outcomes.

The literature review delves into three main areas relevant to AI in science education. Firstly, it explores the body of research focused on the use of AI in the teaching and learning of science subjects. This section discusses the broad applications of AI technologies in classrooms and how these technologies are shaping the way science is taught, particularly in areas requiring complex problem-solving, experimentation, and conceptual understanding. AI tools such as virtual labs and AI-driven simulations allow students to engage with scientific concepts in interactive ways, enhancing their grasp of difficult topics.

Secondly, the review examines studies on the specific AI applications being utilized in science education. These include tools that support adaptive learning, personalized feedback, and automated assessments, all of which aim to cater to diverse learner needs. By focusing on how these applications are integrated into the curriculum, the review highlights best practices, and the challenges teachers face in implementing these tools effectively.

Finally, the literature review investigates research on the effectiveness of AI in improving educational outcomes in science. This includes studies that measure student performance, engagement, and attitude shifts towards science learning when AI tools are employed. The review also considers teacher perspectives on AI integration, particularly in terms of the support and training required to utilize these technologies meaningfully in the classroom.

1.6 Overview of Methodology

This research employs a systematic review methodology to investigate the integration of artificial AI in secondary school science education. The systematic review approach allows for a comprehensive analysis of existing literature, synthesizing findings from multiple studies to identify trends, challenges, and opportunities associated with the use of AI in this educational context.

The review process began with a comprehensive search of relevant databases to identify articles that focus on AI applications in science education. Inclusion criteria were established to select studies that specifically address the effectiveness of AI, teachers' and learners' attitudes towards AI, and various AI applications in teaching and learning. This process ensured that the selected studies were pertinent to the research questions.

Data extraction was conducted using a coding table designed to capture key information from each study, including the year of publication, aims, research design, findings, and conclusions.

This structured approach facilitated the organization and analysis of the literature, allowing for a clearer understanding of the current state of research on AI in science education.

The analysis involved thematic synthesis of the extracted data, identifying common themes and patterns across studies. This method enables the aggregation of findings, highlighting areas of consensus as well as gaps in the literature that warrant further investigation.

Ethical considerations for this systematic review focused on the proper citation of all included studies and ensuring transparency in the selection and analysis processes. No primary data were collected from human subjects, thereby minimizing ethical concerns.

While this study aims to provide valuable insights into the integration of AI in science education, it is important to acknowledge potential limitations inherent to systematic reviews, such as publication bias and the variability in study quality.

Chapter Two

Conceptual Framework and Literature Review

2.1 Introduction

This chapter presents a comprehensive examination of the integration of artificial intelligence in secondary school science education, focusing on the conceptual framework and the literature review. The aim of this chapter is to provide a structured analysis of key components related to AI's role in enhancing science education and to synthesize existing research on this topic.

The conceptual framework establishes the foundational elements that underpin the study. It includes a detailed exploration of AI in education, encompassing its definition, scope, and types. This framework specifically addresses its application within secondary school science subjects, including life sciences, physical sciences, biology, human biology, combined science, integrated science, and chemistry. Furthermore, it examines the preparedness of teachers to integrate AI technologies into their teaching practices and evaluates the effectiveness of AI in fostering student engagement and improving learning outcomes.

In addition to the conceptual framework, the literature review synthesizes current research on the use of AI in teaching and learning science. It aims to answer critical research questions regarding the state of AI integration in education, the prevalent AI applications utilized in science classrooms, and the effectiveness of these applications in enhancing educational outcomes. The literature review is organized into three main subsections: research on the use of AI in teaching and learning of science, research on the use of AI applications in this domain, and research on the effectiveness of AI in science education. Each subsection offers an in-depth analysis of current trends, challenges, and future directions for the use of AI in secondary school science education.

2.2 Conceptual Framework

The integration of Artificial Intelligence (AI) in education has become an increasingly important area of research, particularly in the context of enhancing teaching and learning in secondary school science. AI offers a wide range of applications that can potentially transform science education by improving teaching methodologies, enhancing student engagement, and providing personalized learning experiences. In the regions where there are persistent

challenges related to equitable access to quality education, AI holds promise as a tool that can help bridge gaps in science learning.

This chapter provides a conceptual framework that explores the critical components underlying the integration of AI into secondary school science education. It examines AI's role, scope, and applications in education, with a specific focus on the unique challenges and opportunities within secondary school science classrooms. The chapter also considers the role of teacher preparedness and the effectiveness of AI in delivering science education, along with the various applications used to facilitate science teaching and learning. This framework will guide the systematic review by laying the foundation for analyzing existing literature on AI in science education and its implications for teaching and learning.

2.2.1 Definition and Scope of Artificial Intelligence in Education

AI can be divided into two primary types: General AI and Narrow AI (Fei et al, 2020). Narrow AI focuses on excelling at specific tasks, such as voice assistants like Siri or Alexa, which are adept at understanding and responding to voice commands within defined parameters (Rabelo et al, 2022). Generative AI tools, such as ChatGPT-4 and its precursor, ChatGPT, are currently among the most widely recognized within the educational landscape (Kukulska-Hulme et al., 2022). These tools are engineered to engage users in conversations, responding to commands and generating text that mimics human-produced content rather than machine-generated output (Fei et al, 2020). Their development relies on extensive datasets sourced from the internet, with ChatGPT trained on a corpus of 300 billion words (Kukulska-Hulme et al., 2022). The vastness of these datasets enables the tools to learn predictive patterns, replicate various genres and writing styles, and seamlessly transition between languages (Fei et al, 2020). Moreover, the array of generative AI tools is constantly expanding and evolving. This evolution occurs swiftly, with some tools undergoing changes in nomenclature and structure, while others transition to exclusive, high-cost subscription models.

The applications of AI are diverse and pervasive, encompassing various domains such as virtual assistants, recommendation systems, image recognition, gaming, language translation, chatbots, robotics, and smart home devices (Hwang et al., 2020). Notably, the current selection of generative AI tools suitable for teaching and learning encompasses, but is not limited to, ChatGPT, Machine Learning applications, Intelligent Tutoring Systems, Adaptive Learning Platforms, Virtual Labs and Virtual Reality, and Simulations (Soori et al, 2023). These applications leverage AI algorithms to perform tasks ranging from natural language processing

and image analysis to decision-making and autonomous navigation, thereby enhancing efficiency, convenience, and innovation across numerous industries. Within the sphere of secondary school science teaching, there has been a notable surge in the utilization of artificial intelligence, expanding beyond the traditional perception of AI solely as advanced computing systems, and to encompass embedded computer systems. (Pedro et al., 2019).

2.2.2 Types of AI Applications in Education

Following this expanded understanding of artificial intelligence in secondary school science teaching, it is crucial to explore how it specifically impacts the teaching of sciences. The integration of AI in teaching sciences has the potential to revolutionize pedagogical practices by offering personalized and adaptive learning experiences, optimizing instructional delivery, and enhancing student engagement (Jia et al., 2020). AI-powered educational technologies, such as intelligent tutoring systems, virtual laboratories, and adaptive learning platforms, leverage machine learning algorithms to customize teaching to each learner's requirements, preferences, and learning styles, specifically in the secondary school science teaching sector (Hooda et al., 2022).

AI in education can be categorized based on its application and functionality. Some of the most prominent AI technologies include the following:

a) Simulations and Virtual Reality

Simulations and VR environments allow students to engage in immersive, practical learning experiences that replicate real-world experiments. In science teaching, simulations enable learners to interact with virtual models of complex systems, conduct experiments, and test hypotheses without the need for physical resources (Squire & Jan, 2015). VR further enhances these experiences by immersing students in fully interactive 3D environments, making abstract science concepts more tangible and engaging (Cai et al., 2019).

b) ChatGPT/Generative AI

ChatGPT and similar generative AI models use advanced natural language processing to provide real-time, conversational support for students. These tools can clarify complex scientific concepts, answer questions, and generate educational content such as quizzes, problem sets, or lesson plans. ChatGPT's capacity to simulate human-like conversations creates opportunities for deeper engagement with scientific content (Cooper, 2023).

c) Machine Learning

Machine learning (ML) is a subset of AI that involves systems learning from data and improving over time. In education, ML is used to identify patterns in student learning behaviours and adapt instructional strategies accordingly. ML-based systems help predict student performance, recommend learning pathways, and optimize content delivery based on learners' evolving needs (Zhai, 2021).

d) Intelligent Tutoring Systems

Intelligent Tutoring Systems offer personalized guidance by adapting content and feedback to meet the learner's progress. In science education, ITS helps students tackle difficult concepts such as genetics or physics by providing step-by-step instructions and immediate feedback (Guaman et al., 2017).

e) Adaptive Learning Platforms

These platforms utilize AI algorithms to adjust content delivery based on student progress. Adaptive systems allow science learners to master topics such as chemistry or biology by focusing on areas where they need improvement while advancing more quickly through familiar concepts (Kose & Arslan, 2017).

f) AI-Powered Virtual Laboratories

Virtual laboratories, enhanced by AI, simulate real-world scientific experiments, allowing students to practice hands-on science without the constraints of physical lab environments. These labs provide feedback in real time, guiding students through complex processes such as chemical reactions or biological dissections, while fostering inquiry-based learning (Pyatt & Sims, 2012).

g) Automated Assessment and Feedback Tools

AI-driven assessment tools automatically grade assignments and offer feedback to students. These systems are particularly useful in science education, where they can assess written explanations, problem-solving steps, or coding solutions and provide tailored feedback to promote deeper learning (Murphy, 2019).

In the broader scope of science education, AI enables learners to grasp difficult concepts through adaptive content delivery, simulations, and interactive environments. The use of AI in science classrooms allows students to explore scientific phenomena in ways that would be

impossible or impractical in traditional settings. It also allows teachers to track progress, diagnose learning difficulties early, and provide targeted interventions that enhance conceptual understanding and scientific inquiry (Hwang et al., 2021). The potential for AI to democratize science education and close the equity gap is significant, as these technologies provide access to high-quality learning resources regardless of location or socioeconomic status.

2.2.3 Secondary School Science Teaching: Scope and Subject Areas

Secondary school science teaching involves the structured and systematic instruction of scientific concepts and practices to students typically between the ages of 15 and 19. The goal is to develop learners' understanding of the natural world, equip them with problem-solving skills, and foster scientific thinking. Science education at this level spans various subject areas, each with specific curricula, but collectively aiming to prepare students for higher education, careers in STEM fields, and informed citizenship.

2.2.3.1 Key Subject Areas in Secondary School Science

This research focuses exclusively on secondary school science subjects, such as biology, chemistry, and physics, and deliberately excludes disciplines like engineering, mathematics, and information technology, including CAT (Computer Applications Technology). The science subjects under consideration are those typically taught within secondary school curricula and are fundamental to science education.

Life Sciences/Biology/Natural Sciences

Life Sciences (often called Biology) focuses on the study of living organisms, their structures, functions, and interactions with each other and the environment. It covers genetics, evolution, ecosystems, anatomy, physiology, and cellular biology. Natural Sciences is a broad category that often includes aspects of both Life Sciences and Physical Sciences, with an emphasis on understanding the physical world and natural phenomena. This subject area includes environmental science, ecology, geology, and other earth sciences. AI tools such as virtual dissections, genetics simulators, and intelligent tutoring systems have enhanced biology education by making abstract concepts more accessible and providing interactive learning experiences.

Physical Sciences

Physical Sciences, very often encompass physics and chemistry, dealing with non-living matter and the fundamental principles governing energy, forces, and the behaviour of matter. AI integration in these fields includes simulations of chemical reactions, physics models demonstrating the laws of motion, and AI-driven problem-solving platforms that help students understand difficult equations or scientific phenomena.

Human Biology

Human Biology is a subset of Life Sciences focusing on the anatomy, physiology, and health sciences related to humans. It covers systems of the body, disease, genetics, and human development. AI applications in human biology education include virtual models of the human body, interactive simulations for understanding organ systems, and personalized learning platforms that provide targeted feedback on complex topics.

Combined Science/Integrated Science

These terms are often used to describe curricula that blend aspects of different scientific disciplines into a cohesive study. Integrated Science, for instance, combines chemistry, physics, biology, and earth sciences, teaching students to approach scientific problems holistically. AI applications here include interdisciplinary simulations that require students to draw on multiple fields of knowledge, adaptive learning systems that track progress across subjects, and AI-based assessments that cover a range of scientific domains.

Chemistry

Chemistry focuses on the study of matter, its properties, how substances interact, and the reactions they undergo. AI applications in chemistry education include virtual labs where students can safely perform chemical reactions, AI-powered tools for molecular visualization, and adaptive systems that guide learners through stoichiometry, bonding, and thermodynamics problems.

2.2.4 Impact of AI on Science Teaching Methods

The integration of AI in secondary school science teaching has significantly influenced various teaching methods, making the instruction of scientific concepts more interactive, personalized, and data driven (Baidoo-Anu & Ansah, 2020). One such method is inquiry-based learning, which encourages students to explore scientific problems, ask questions, and conduct

experiments to find solutions (Holbrook & Kolodner, 2013). AI enhances this method through intelligent tutoring systems that guide learners in problem-solving, provide hints, and offer feedback during experiments or tasks (Hao et al., 2024). AI-driven virtual labs also enable students to conduct safe experiments with real-time feedback, fostering a deeper understanding of the scientific process.

Project-based learning (PBL), another method impacted by AI, allows students to work on complex, real-world problems over extended periods (Krajcik & Czerniak, 2018). AI supports this approach by providing simulations, data analysis tools, and collaborative platforms where students can work together on science projects, often without being constrained by physical lab resources (Kim, 2022). In the flipped classroom model, where instructional content is delivered outside of class, AI tools assess student understanding and adjust content delivery. These platforms track student progress through pre-class tasks and suggest areas of improvement during in-class activities.

AI also plays a significant role in personalized learning pathways. AI-powered platforms adapt content delivery based on students' unique strengths, weaknesses, and learning paces (Berland et al., 2015). This allows for differentiated instruction, particularly useful in heterogeneous science classrooms where learners may have varying levels of prior knowledge (Jia et al., 2023). Additionally, AI tools facilitate collaborative learning by providing virtual platforms where students can work together on scientific problems, share data, and engage in collective inquiry. Some AI-driven systems even monitor group dynamics and offer feedback on collaboration efficiency, improving teamwork and communication (Holstein et al., 2019).

Lastly, AI-powered assessment systems offer real-time feedback on student performance, enabling continuous formative assessment. In science education, these tools can assess both theoretical knowledge, such as answering multiple-choice questions on biological processes, and practical skills, like conducting virtual chemistry experiments, offering immediate insights for improvement (Ivanović et al., 2022). Overall, the inclusion of AI in these teaching methods has revolutionized secondary science education, making learning more dynamic, personalized, and immersive, ultimately enhancing students' understanding of scientific concepts.

2.2.5 Teacher Preparedness and AI Integration

The successful integration of AI in secondary school science education relies heavily on teacher preparedness. This encompasses not only teachers' familiarity with AI technologies but also

their ability to effectively incorporate these tools into their teaching practices. Teacher preparedness can be broadly categorized into professional development, attitudes toward AI, and the challenges they face in adapting to AI technologies.

Professional Development for AI Integration

Professional development is a critical factor in ensuring that teachers are equipped to integrate AI into their classrooms. Effective professional development programs offer hands-on training in AI tools, strategies for integrating AI into lesson planning, and opportunities for continuous learning. For instance, workshops on the use of AI-driven virtual laboratories, AI-based simulations, or adaptive learning platforms provide teachers with practical knowledge that they can implement in their teaching (Luan et al., 2020).

In many regions, the lack of structured and widespread professional development on AI integration has been a challenge. Many teachers report needing more in-depth training on how to use AI tools, especially in under-resourced schools where access to technology is limited. This professional development must not only focus on technical skills but also on pedagogical strategies for effectively incorporating AI to enhance student learning outcomes.

Teachers' Attitudes Toward AI

Teacher attitudes towards AI play a significant role in how successfully these tools are adopted. Research indicates that teachers' willingness to engage with AI is influenced by their comfort level with technology, their belief in AI's potential to improve learning, and their confidence in using AI tools. Some educators view AI as a valuable resource that can help streamline administrative tasks, personalize learning, and make complex scientific concepts more accessible (Galindo-Domínguez et al., 2024). However, others may feel apprehensive about its potential to replace traditional teaching methods or even their role as educators.

In many cases, positive attitudes toward AI increase when teachers see tangible benefits, such as improved student engagement or enhanced learning outcomes. On the other hand, resistance may arise due to concerns about AI's perceived complexity or scepticism regarding its impact on education. Thus, fostering positive attitudes through professional support and demonstrating the real-world benefits of AI tools can encourage teachers to integrate AI more effectively into their classrooms.

Challenges in Adapting to AI Technologies

Teachers often face challenges when adapting to AI technologies, especially in regions with limited resources. In South Africa, disparities in access to technology can hinder the integration of AI in science teaching, as schools in under-resourced areas may lack the infrastructure necessary to support AI tools. Additionally, teachers may struggle with time constraints, as the demands of traditional teaching can leave little room for learning and implementing new technologies (Hodgkinson-Williams & Arinto, 2017).

Another challenge is ensuring that AI tools are aligned with the existing science curriculum. Teachers must not only learn how to use AI but also how to integrate it in ways that meet curriculum objectives and enhance student understanding of scientific concepts. Overcoming these challenges requires a combination of systemic support, tailored professional development, and access to the necessary technological resources.

2.2.6 Effectiveness of AI in Science Education

The effectiveness of AI integration in secondary school science education is determined by several key outcomes, including improved student engagement, personalized learning experiences, and enhanced understanding of complex scientific concepts. These outcomes are typically measured through metrics such as student performance, teacher efficiency, and learner engagement.

Improved Student Engagement

AI has the potential to significantly increase student engagement in science education by offering interactive and immersive learning experiences. Tools like simulations, virtual reality, and AI-powered tutoring systems provide students with opportunities to explore scientific phenomena in ways that traditional teaching methods may not allow. For example, virtual labs can simulate chemical reactions or biological processes, allowing students to experiment in a risk-free environment (Holstein et al., 2019). Furthermore, studies have shown that students are more likely to be motivated and engaged when using AI-driven tools, as these technologies make learning more interactive and visually stimulating. Gamified learning environments, for example, can make science lessons more enjoyable, leading to increased participation and a deeper interest in the subject matter (Luckin et al., 2016).

Personalized Learning

One of the most significant advantages of AI in education is its ability to provide personalized learning experiences (Ledesma & García, 2017). AI-powered adaptive learning platforms tailor instructional content to meet individual learners' needs, addressing gaps in knowledge and pacing lessons according to each student's progress. This is particularly beneficial in science education, where students often struggle with varying levels of understanding of complex concepts (Hellings & Haelermans, 2020). For instance, AI-based platforms can adjust the difficulty of quizzes, provide targeted feedback, and suggest additional resources based on a student's performance. This personalized approach not only helps students grasp difficult topics but also empowers them to take ownership of their learning process (Woolf, 2020). Personalized learning also frees teachers from having to cater to every individual need, allowing them to focus on more critical instructional activities.

Enhanced Understanding of Complex Concepts

AI tools, especially those that offer visualizations and simulations, are particularly effective in helping students understand complex scientific concepts. For example, in biology, AI-powered simulators can model cellular processes or the intricacies of ecosystems, making abstract concepts more tangible. Similarly, in physics and chemistry, AI-driven visualizations can demonstrate the laws of motion or molecular structures, providing learners with a more comprehensive understanding of how scientific principles work (Cooper et al., 2020). Therefore, by providing students with multiple ways of interacting with scientific material, whether through text, visuals, or hands-on simulations, AI enhances cognitive understanding and allows learners to experiment with and apply knowledge in various contexts.

Metrics of Effectiveness: How the effectiveness is measured

Several metrics are commonly used to measure the effectiveness of AI in secondary school science education such as student performance, teacher efficiency and learner engagement. Improved test scores and higher rates of concept retention are key indicators of the effectiveness of AI tools in education. Studies have shown that students who use AI-based learning systems perform better on assessments and demonstrate a deeper understanding of scientific concepts (Wang et al., 2011). Moreover, AI tools can reduce the administrative workload for teachers, allowing them to focus more on instructional time and direct engagement with students. Tools that provide automated grading, personalized feedback, and classroom analytics have been shown to enhance teacher productivity (Chen et al., 2020).

Increased student participation, engagement with interactive content, and motivation to learn science are other key indicators. AI tools that provide real-time feedback, personalized learning pathways, and interactive simulations foster a more engaging learning environment (Blumenfeld et al., 2019).

In conclusion, the interaction between these key concepts, AI applications, teacher preparedness, and AI effectiveness forms the foundation of this study. However, the success of these tools is closely linked to the preparedness of teachers, who play a crucial role in implementing AI in ways that are pedagogically sound, and curriculum aligned.

Teacher preparedness directly influences the effectiveness of AI integration. Teachers who receive proper professional development and feel confident in using AI are more likely to successfully integrate these technologies into their teaching. On the other hand, teachers who lack the necessary training or have negative attitudes toward AI may struggle to effectively implement these tools. Furthermore, the effectiveness of AI can be measured by improved student outcomes, including better understanding of scientific concepts, higher engagement levels, and enhanced performance on assessments. When AI is effectively integrated and supported by well-prepared teachers, it can transform secondary school science education, offering students more personalized, engaging, and effective learning experiences.

2.3 Literature Review

The integration of Artificial Intelligence in secondary school science education is a rapidly evolving field that promises to enhance teaching methodologies and improve student learning experiences. As educational technology advances, it is crucial to understand how AI can support educators and learners in achieving educational goals. This literature review addresses the following research questions: What is the current state of research on the use of AI in teaching and learning science? What are the prevalent AI applications utilized in this domain? How effective are these applications in enhancing educational outcomes?

AI in education encompasses various technologies and methodologies aimed at personalizing and optimizing learning processes. An interdisciplinary field, often referred to as Artificial Intelligence in Education (AIEd), aims to leverage AI tools to enrich instructional practices across different academic disciplines (Chen et al., 2020). In science education, AI has the potential to transform traditional pedagogical approaches, offering personalized learning experiences, adaptive feedback, and innovative assessment methods. Previous studies have

highlighted numerous AI-driven technologies, such as intelligent tutoring systems, adaptive learning platforms, and virtual laboratories, which can facilitate engaging and effective learning environments.

Despite the promising applications of AI, concerns regarding ethical implications, data privacy, and the balance between AI-driven instruction and traditional teaching methods have been raised. These challenges underscore the need for a thorough examination of the existing literature on AI's integration into science education to understand its benefits and drawbacks.

This literature review is structured around three main subsections: research on the use of Artificial Intelligence in teaching and learning of science, research on the use of Artificial Intelligence applications in teaching and learning of science, and research on the effectiveness of AI in teaching and learning of science**.** Each section aims to provide a comprehensive analysis of current trends, challenges, and future directions for AI in secondary school science education.

2.3.1 Research on the Use of Artificial Intelligence in Teaching and Learning of Science

The integration of AI into teaching and learning processes has garnered increasing attention within the educational landscape, particularly in the domain of science education. Research in this area reveals a multifaceted understanding of how AI can be harnessed to improve educational experiences, specifically in secondary school science classrooms.

Studies indicate that AI can serve as a powerful tool in facilitating personalized learning experiences for students. For instance, intelligent tutoring systems are designed to adapt to individual student needs, providing tailored feedback and support in mastering complex scientific concepts (Darayseh, 2023; Hao et al., 2024). By employing algorithms that assess students' performance and learning patterns, ITS can identify knowledge gaps and offer resources suited to the learner's level (Okunade, 2024). This personalized approach is crucial in a diverse classroom setting, where students possess varying degrees of understanding and learning styles.

Moreover, the utilization of AI-driven platforms enables educators to streamline administrative tasks, allowing them to focus more on instruction and individualized student support (Pedro et al., 2019). For example, AI systems can automate grading and provide immediate feedback on assignments, enhancing the efficiency of assessment processes (Murphy, 2019). This integration not only alleviates the workload of teachers but also fosters a more responsive

learning environment for students, promoting active participation and engagement with scientific content.

Despite the numerous benefits associated with AI in science education, concerns regarding ethical implications and data privacy must be addressed. Research highlights the risks of over-reliance on AI, which may undermine the role of educators in cultivating critical thinking and creativity among students (Chassignol et al., 2018). Furthermore, algorithmic biases can disproportionately affect underrepresented student groups, raising equity issues within educational settings (Hooda et al., 2022). As such, it is essential for educators and policymakers to consider these challenges in the implementation of AI technologies in science teaching.

In summary, the existing body of research underscores the potential of AI to transform teaching and learning in secondary school science education. While AI offers innovative solutions for personalized learning and efficient assessment, ethical considerations and equity issues warrant careful attention. Further systematic reviews are necessary to assess the comprehensive implications of AI integration in secondary school science teaching, ensuring that these technologies are utilized effectively and inclusively.

2.3.2 Research on the Use of Artificial Intelligence Applications in Teaching and Learning of Science

The application of AI in secondary school science education encompasses a wide range of technologies designed to enhance the instructional process and improve student learning outcomes. Various AI applications have emerged, providing innovative approaches that foster engagement and deeper understanding of scientific concepts.

One notable development in this field is the integration of conversational AI models, such as ChatGPT, into the classroom environment. Cooper (2023) explored the efficacy of using ChatGPT in a secondary school science context, demonstrating its potential to serve as an interactive educational tool. The study revealed that ChatGPT can provide personalized explanations, address student queries, and facilitate discussions on scientific topics. By leveraging natural language processing capabilities, this AI model allows students to engage in dynamic conversations, promoting active learning and enhancing their comprehension of complex scientific ideas.

ITS have also gained prominence as effective AI applications in science education. Research indicates that ITS tailored to secondary school science curricula can offer personalized

feedback and adaptive instruction, supporting students in mastering challenging scientific concepts (Hao et al., 2024). These systems assess individual student performance and dynamically adjust instructional content, fostering personalized learning experiences that cater to diverse learning needs.

Additionally, adaptive learning platforms customized for secondary school science subjects enable educators to create a more personalized educational environment. These platforms analyse student data to adjust instructional content and pacing based on individual performance, thereby enhancing the learning experience (Muhammad et al., 2023; Hooda et al., 2022). Such AI-driven technologies empower students to engage with the material at their own pace, allowing for deeper exploration and understanding of scientific concepts.

Virtual labs and simulations are another significant application of AI in science education. These tools allow students to conduct experiments and explore scientific phenomena in a virtual environment, overcoming limitations often associated with physical laboratory settings (Rajaratnam, 2023; George and Kolobe, 2014). Research demonstrates that virtual labs enhance students' inquiry-based problem-solving skills and foster a more interactive learning experience, ultimately leading to improved scientific literacy (Lestari and Supahar, 2020; Pyatt & Sims, 2012).

Despite the potential benefits of AI applications in science education, several challenges must be addressed. Issues related to data privacy and security arise, as AI systems often require access to sensitive student information (Chassignol et al., 2018). Furthermore, algorithmic bias may affect the equitable distribution of learning opportunities, particularly for underrepresented groups (Pedro et al., 2019). As such, it is crucial for educators and stakeholders to approach the implementation of AI technologies in secondary school science education with a critical lens, ensuring that ethical considerations are prioritized.

In conclusion, the integration of AI applications in secondary school science education presents a transformative opportunity for enhancing teaching and learning processes. From intelligent tutoring systems and adaptive learning platforms to virtual labs, AI technologies are poised to revolutionize the educational landscape. However, addressing ethical concerns and ensuring equitable access to these technologies remains paramount to their successful integration in science classrooms.

2.3.3 Research on the Effectiveness of AI in Teaching and Learning of Science

The effectiveness of AI in enhancing teaching and learning outcomes in secondary school science education has become a focal point for researchers and educators alike. As AI technologies continue to permeate educational settings, various studies have sought to evaluate their impact on student engagement, learning outcomes, and overall instructional effectiveness.

Numerous studies have reported significant improvements in student engagement attributed to the integration of AI in science education. For instance, Darayseh (2023) and Lestari and Supahar (2020) highlight that AI-driven platforms facilitate interactive learning experiences that capture students' attention and sustain their interest in complex scientific concepts. By providing personalized feedback and adaptive learning paths, AI tools foster a more engaging learning environment, encouraging students to take an active role in their education (George and Kolobe, 2014).

Research has also demonstrated that AI applications can lead to improved academic performance in science subjects (Lee and Perret, 2022). Intelligent tutoring systems have shown efficacy in providing tailored instruction and support, enabling students to master challenging topics more effectively (Guaman et al., 2017; Ahmad et al., 2021). The ability of ITS to assess individual student progress and adapt instructional strategies accordingly enhances learning outcomes, particularly for students who may struggle in traditional classroom settings (Hao et al., 2024).

Additionally, the use of AI in virtual laboratories has been linked to increased scientific literacy among students. By simulating real-world scientific experiments, virtual labs allow learners to explore concepts in an interactive and safe environment (Nja et al., 2023; Pyatt & Sims, 2012). This experiential learning approach enables students to develop critical thinking and problem-solving skills, which are essential for success in the field of science. Rajaratnam (2023) further emphasizes that virtual labs can bridge the gap between theoretical knowledge and practical application, reinforcing students' understanding of complex scientific principles.

Despite the promising findings regarding the effectiveness of AI in science education, it is essential to acknowledge the challenges that accompany its implementation. Issues such as data privacy, algorithmic bias, and the potential for over-reliance on technology present significant concerns (Chassignol et al., 2018; Pedro et al., 2019). Educators must ensure that AI tools are utilized ethically and inclusively, fostering an equitable learning environment for all students.

In summary, the research indicates that AI has the potential to enhance the effectiveness of teaching and learning in secondary school science education. By improving student engagement, facilitating personalized learning experiences, and promoting critical thinking skills, AI applications can significantly impact educational outcomes. However, careful consideration of ethical implications and potential challenges is crucial to maximize the benefits of AI integration in science classrooms.

2.3.4 Future Trends in AI and Secondary School Science Teaching

Looking ahead, AI is set to play an increasingly prominent role in secondary school science teaching, with several emerging trends shaping the future of teaching and learning (Luan et al, 2020). One key trend is enhanced learning analytics, where AI-driven data analysis tools offer detailed insights into student performance and learning patterns (Luan et al, 2020). These insights allow teachers to customize their instruction to meet the needs of individual students, leading to more effective and personalized teaching methods.

Another trend is AI-driven collaborative platforms, which support teamwork and collaboration among students (Ivanović et al, 2022). These platforms enable learners to work together on projects, share resources, and engage in discussions, fostering a deeper understanding of scientific concepts through collaborative learning experiences. This approach encourages communication and problem-solving skills (Ivanović et al, 2022), which are essential in secondary school science teaching.

Finally, AI in assessment is gaining traction, with automated grading and feedback systems becoming more prevalent (Murphy, 2019). These systems can quickly and accurately evaluate student work, reducing the burden on teachers and providing more immediate feedback to students. This allows educators to focus more on instruction and personalized support, creating a more efficient and responsive educational environment. Together, these trends indicate a future where AI plays a central role in enhancing secondary school science teaching, offering new tools and approaches that can improve learning outcomes and streamline teaching processes.

Summary of the Chapter

This chapter has provided a comprehensive exploration of the conceptual framework and literature review regarding the integration of artificial intelligence in secondary school science education. The discussion began with the establishment of a conceptual framework that defined

key concepts related to AI, including its scope and various applications within the context of secondary school science subjects. It highlighted the importance of teacher preparedness and the potential effectiveness of AI technologies in enhancing educational practices and learning outcomes.

The literature review synthesized existing research on the use of AI in science education, organized into three main subsections: research on the use of AI in teaching and learning of science, research on AI applications in this domain, and research on the effectiveness of AI in improving educational outcomes. Through this synthesis, the review illuminated the multifaceted role of AI in transforming traditional pedagogical practices, facilitating personalized learning experiences, and addressing the diverse needs of students in science classrooms.

Key findings from the literature indicate that while AI presents significant opportunities for enhancing engagement and understanding of complex scientific concepts, ethical considerations, data privacy concerns, and potential biases must be addressed to ensure equitable access and implementation. The insights gained from this chapter will serve as a foundational basis for the subsequent chapters of this research project, guiding the investigation into the integration of AI technologies in secondary school science education and their impact on teaching and learning.

Chapter Three

Research Methodology

3.1 Introduction

This study employs a mixed-method systematic review approach, which integrates both qualitative and quantitative data to provide a comprehensive analysis of the integration of AI in secondary school science education. A mixed-method approach is particularly suitable for this research as it allows for the synthesis of diverse types of data, ensuring a more nuanced understanding of how AI impacts teaching and learning. As Creswell and Creswell (2011) and Muza (2014) suggest, a research method encompasses the planning, design, data collection, and analysis strategies that researchers use to carry out an investigation (Creswell and Creswell, 2011; Muza, 2014). The mixed-method approach is grounded in the idea that combining qualitative and quantitative evidence offers deeper insights into complex phenomena, enabling a more holistic evaluation of both measurable outcomes and contextual factors (Creswell & Plano Clark, 2018).

In the context of this systematic review, the mixed-method approach is appropriate because the available literature on AI in secondary school science education includes both quantitative studies (e.g., student performance metrics, effect sizes) and qualitative research (e.g., teacher experiences, implementation challenges). By incorporating both types of data, this study addresses the need for a comprehensive understanding of AI's effectiveness in diverse educational contexts. Mixed-method research is especially valuable in emerging fields like AI in education, where the integration of qualitative insights can illuminate the underlying dynamics of AI implementation, while quantitative data can confirm the tangible effects on student learning outcomes (Johnson, Onwuegbuzie, & Turner, 2007).

This systematic review rigorously collects, analyses, and synthesizes studies from a range of contexts, providing both an overarching view of trends and a detailed examination of specific outcomes. A systematic review consolidates research findings, allowing for an unbiased evaluation of AI's impact on science education, particularly as the literature on AI in education continues to evolve. By combining qualitative and quantitative data, this mixed-method review offers a more complete picture of AI's role in enhancing secondary school science teaching, as well as identifying gaps and areas for future research (Tashakkori & Teddlie, 2010).

This chapter will detail the data sources, search strategies, and inclusion/exclusion criteria used to identify relevant studies, ensuring a broad yet focused exploration of the topic. Additionally, the study selection process will be explained, along with the frameworks used to assess the quality and reliability of the included studies, such as the Critical Appraisal Skills Programme (CASP) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). CASP provides specific checklists for various types of research designs, which were used to evaluate the methodological rigor of each study (Long et al., 2020). Studies that did not meet the minimum quality threshold were excluded from the review. These steps are essential to ensure that the conclusions drawn are based on high-quality, credible research that provides valuable insights into the role of AI in secondary school science education.

3.2 Systematic Review Methods/ Strategies

3.2.1 Data Sources

To conduct a comprehensive systematic review on the integration of AI in secondary school science teaching, I utilized a rigorous search strategy across multiple academic databases and platforms. My primary source of data was the Web of Science (WoS), which is known for its extensive and high-quality coverage of scientific research across various disciplines. Additionally, I used Scopus as a secondary source, as it offers a broad database with bibliographic information from major publishers, including Elsevier, Springer, ACM, and IEEE. These databases provided a robust foundation for gathering relevant studies. In addition to WoS and Scopus, I explored other significant sources to ensure comprehensive coverage. ResearchGate was used to access a broader network of researchers and find additional publications, including conference papers and preprints. PubMed, a database specializing in life sciences and biomedical topics, was also consulted for relevant research in secondary school science teaching and technology integration. Furthermore, I utilized the University of the Witwatersrand's (Wits) online library to access academic journals, dissertations, and other scholarly resources. This provided a diverse range of sources for my review.

Given the varied applications of AI in secondary school science teaching, my search strategy encompassed a wide range of publication types, including journal articles, conference papers, and book chapters, without restricting to specific journals. I focused on studies published between 2000 and 2024, providing a comprehensive view of the evolving landscape of AI in secondary school science teaching.

3.2.2 Search Strategies for Articles

To find relevant studies, I developed a set of targeted search strategies that included multiple keyword categories, ensuring comprehensive coverage of the topic. First, I used keywords related to AI and specific applications, such as "artificial intelligence," "AI," "AIED," "machine learning," "intelligent tutoring system," "adaptive learning," "intelligent agents," and similar terms. Second, I incorporated keywords focused on secondary school science teaching, such as "science," "education," "teaching," and "learning." To broaden the scope, I also included phrases like "STEM," "physical sciences," "biology," "life sciences," and "science teaching." These terms helped cover the various subjects where AI is applied in educational contexts.

Third, I used terms indicating the education level, focusing on "secondary school," "middle school," and "high school." I also excluded terms related to higher education or early childhood education to maintain the focus on secondary school science teaching. This approach allowed me to target research studies specifically relevant to this level of education. Additionally, to ensure comprehensive coverage, I considered other relevant terms, including "STEM education," "science curriculum," "pedagogy," "education technology," and "digital learning." By incorporating a wide range of keywords, the search strategy effectively covered the key themes and applications associated with AI in secondary school science teaching.

3.2.3 Inclusion and Exclusion Criteria

To ensure the quality and relevance of the articles included in my review, I followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. I used the PRISMA flowchart to map out the identification, screening, inclusion, and exclusion process for articles. Initially, I scanned titles and abstracts to determine relevance. After this preliminary review, I conducted a detailed examination of the full text to confirm each article's suitability for inclusion in the systematic review.

The inclusion criteria were based on the following factors:

Table 1: Inclusion and exclusion criteria for article selection.

Criterion	Inclusion	Exclusion
Article Topic	Integration of AI in secondary school science teaching	Do not integrate AI into secondary school contexts.
Article Type	Empirical/Non-Empirical Studies.	Editorials, Reviews, Commentaries, Conference abstracts, and Opinion papers.
Article Publication	Published.	Unpublished or pre-printed or under review.
Year of Publication	From Year 2000 to 2024	Before Year 2000
Article Availability	Available online as full text.	Not available online.
Article Language	English	Non-English
Population	Grade 10 - 12 settings or within the age range of 15-20.	K-9, Higher and adult education are excluded.
School subjects	Life Sciences/Biology, Physical Sciences and Chemistry	Mathematics, Engineering, Information Technology and CAT subjects
Measure	Studies related to AI-enabled science pedagogy in secondary school that discuss learning tools, pedagogical strategies, learning outcomes, and assessment methods.	The focus is not related to AI in secondary school science teaching and learning context.

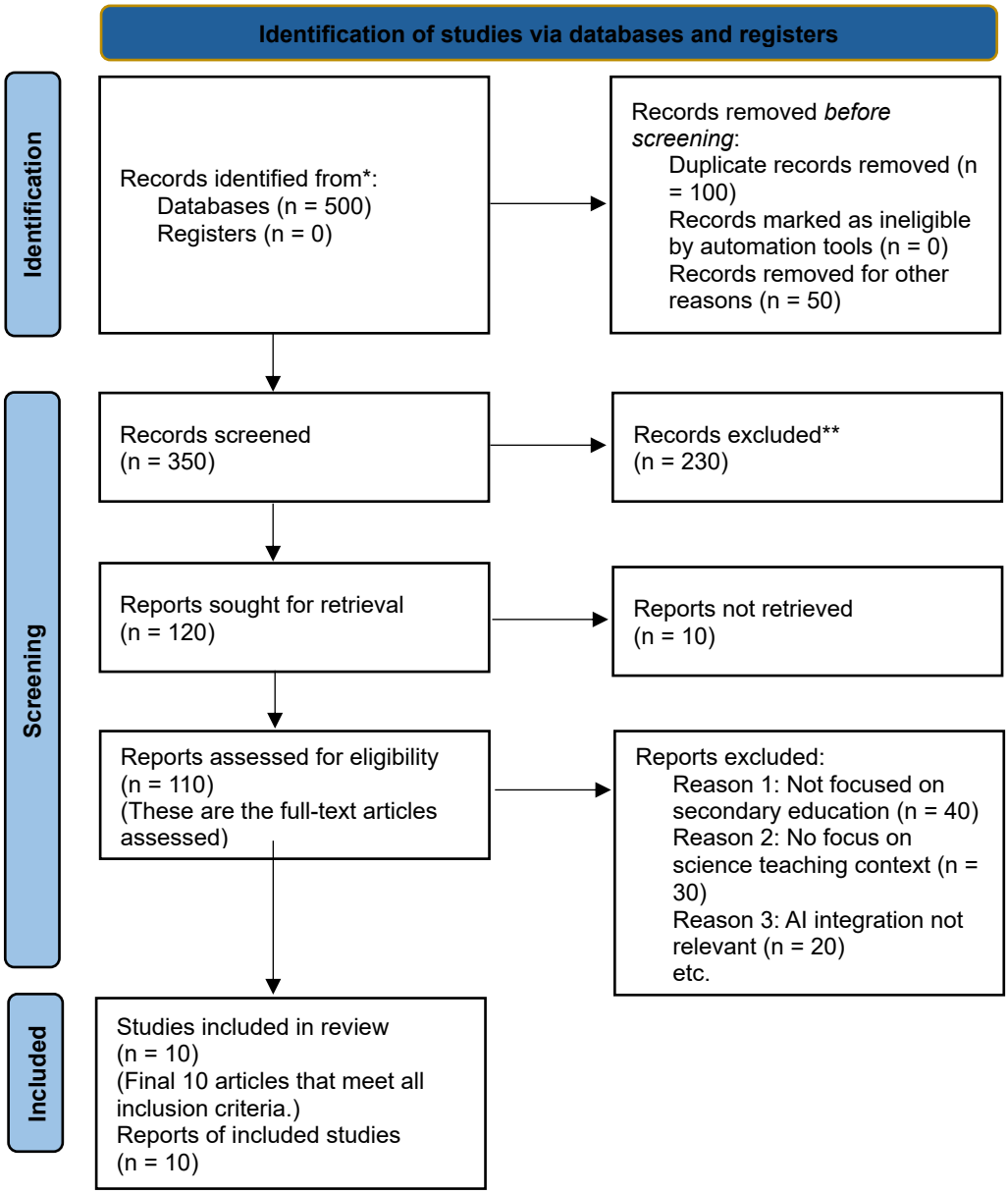
3.2.4 Study Selection and Screening Process

The study selection process involved a multi-step approach. Initially, I conducted an initial screening of titles and abstracts to identify potentially relevant studies. This was followed by a detailed examination of the full texts to confirm the suitability of each study for inclusion in the systematic review.

The following PRISMA flow diagram outlines the systematic process used to identify, screen, and include studies in this review. It visually represents the stages of study selection, including the total number of records identified, the number of studies excluded at various stages, and

the final number of studies included in the review. This diagram ensures transparency in the selection process, highlighting how the reviewed studies were narrowed down to meet the inclusion criteria.

Figure 1: PRISMA flow diagram



3.2.5 Assess Quality of Included Studies

To ensure the quality and reliability of the included studies, I followed a structured approach to assess each study's quality. This involved evaluating the studies based on specific criteria such as methodological rigor, sample size, reliability, validity, and relevance to the research topic. For instance, I examined whether studies employed control groups, the appropriateness of the chosen methodologies, and how clearly the research questions were articulated. I used established frameworks like the Critical Appraisal Skills Programme (CASP) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) to guide the quality assessment process. Each study was scored based on these criteria, and those that did not meet the minimum quality standards were excluded from further analysis. Based on my initial assessment, I expect that some studies may demonstrate limitations, such as small sample sizes or insufficient methodological rigor, which could affect the reliability of their findings.

In addition to assessing the quality of individual studies, I also considered the overall quality and consistency of the body of research. This included identifying potential biases, examining the range of study designs, and ensuring a balanced representation of perspectives within the included studies. The quality assessment process was an essential step in ensuring that the conclusions drawn from this systematic review were based on high-quality, reliable research that provides a solid foundation for further exploration and application in the field of AI in secondary school science teaching.

Table 2: CASP Qualitative Research Checklist

Author(s) & Year	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Darayseh (2023)	y	y	y	y	y	y	y	y	y	100%
Nja et al. (2023)	y	y	y	y	y	y	y	y	y	100%
Bogusevschi et al (2020)	y	n	y	y	y	unclear	y	y	y	80%
Okunade (2024)	y	y	y	y	y	y	y	y	y	100%
Cooper (2023)	y	y	y	y	y	y	y	y	y	100%
Park et al. (2023)	y	y	y	y	y	y	y	y	y	100%
Gambari et al (2018)	y	n	y	y	y	unclear	y	y	y	80%
Hao (2024)	y	n	y	y	y	unclear	y	y	y	80%
Lestari & Supahar (2020)	y	y	y	y	y	y	y	y	y	100%
George & Kolobe (2014)	y	y	y	y	y	y	y	y	y	100%
Lee & Perret (2022)	y	n	y	y	y	y	y	y	y	90%

CASP Questions: Q1-Q10

1. Was there a clear statement of the aims of the research?
2. Is a qualitative methodology appropriate?
3. Was the research design appropriate to address the aims of the research?
4. Was the recruitment strategy appropriate to the aims of the research?
5. Was the data collected in a way that addressed the research issue?
6. Has the relationship between the researchers and participants been adequately considered?
7. Have ethical issues been taken into consideration?
8. Was the data analysis sufficiently rigorous?
9. Is there a clear statement of findings?
10. How valuable is the research?

3.2.6 Rationale for Time Frame Selection

The time frame from 2000 to 2024 was selected for this systematic review to ensure comprehensive coverage of studies that reflect the evolution of AI applications in education. The year 2000 was chosen as a starting point because it marks a significant period of technological advancement, with rapid developments in AI tools and educational technologies becoming more prevalent. By extending the review up to 2024, this research captures the most recent studies, ensuring that emerging trends and the latest AI innovations in secondary school science education are included. This period allows for an exploration of how AI's role in science education has developed over two decades, including the early adoption of AI and its more sophisticated integration in recent years.

3.3 Analysis of the Studies

The analysis of the studies included in this systematic review uses a structured approach to synthesize both qualitative and quantitative data from the selected articles. The goal is to categorize, compare, and critically assess the methodologies, AI applications, and educational outcomes discussed across the 10 studies.

3.3.1 Coding of studies

To systematically analyse the studies, a coding framework was developed based on key research components, such as the year of publication, research design, theoretical framework, and findings. This coding process was crucial in identifying patterns, comparing methodologies, and synthesizing the results across diverse studies. The coding framework enabled a structured analysis by ensuring consistency in how each study was interpreted and categorized.

Each study was reviewed and coded according to the predefined columns in the Coding Studies Framework (Table 3), allowing for a detailed comparison of different aspects, such as the aims and objectives of the studies, the AI applications used, and the professional development models referenced. This systematic coding process allowed for both descriptive and analytical insights.

Descriptive coding involved recording basic information, such as the study's research design (e.g., qualitative, quantitative, mixed methods), sampling, and key findings. This step helped in organizing the studies to map out the range of AI applications and research contexts.

Following this, a thematic analysis was conducted based on patterns emerging from the coded data. For example, by comparing the findings and conclusions across studies, we could identify recurring themes, such as the impact of AI on personalized learning, the challenges of teacher preparedness, and the effectiveness of AI applications in different science subjects.

The coding framework also facilitated a cross-comparative analysis, enabling the grouping of studies according to their focus areas (e.g., effectiveness of AI, teacher preparedness, and student engagement). This allowed for the identification of key trends in AI integration in science education, as well as gaps in the literature where further research is needed.

By systematically coding each study's theoretical and conceptual framework, we were also able to assess the underlying assumptions guiding AI implementation. This helped in understanding how the studies positioned AI within broader educational theories and how that influenced the outcomes reported.

Table 3: Coding of the Studies Included in the systematic review

Year	Article Title	Aims/Objectives/Questions	Theoretical & Conceptual Framework (T&C Framework)	Research Design (R/Design)	Sampling	Professional Development (PD) Model	Evaluation Criteria	Findings	Conclusions/Recommendations
2024	Empirical Research on AI Technology-Supported Precision Teaching in High School Science Subjects (Hao et al.)	To analyse the effectiveness of an AI-supported precision teaching model in improving students' academic performance in science subjects	Precision Teaching, Behavioural Learning Theory, Precision Teaching 2.0, Big Data, AI-supported learning	Experimental study with intervention and control groups	Nine classes (540 students) from a high school in China; six intervention classes and three control classes	Teacher training on precision teaching model and AI integration	Academic performance improvement (test scores), self-directed learning behaviors	Significant improvement in students' academic performance, especially in mathematics and chemistry; increased self-directed learning and positive attitudes toward science	Recommend further implementation of AI technology in teaching models to enhance efficiency and reduce teacher burdens.
2023	Examining Science Education in ChatGPT: An Exploratory Study of Generative Artificial Intelligence (Cooper)	Explore how ChatGPT answers science education questions and potential applications for educators	Constructivist Learning Theory, Generative AI in education	Exploratory self-study methodology	Various science education contexts involving AI	Reflection on the use of AI tools for curriculum design and assessment	Alignment of ChatGPT responses with key themes in science education research	ChatGPT can enhance resource creation but raises concerns about accuracy and ethical implications; promotes critical thinking in education	Encourage responsible use of AI, prioritizing critical thinking and adaptation of AI-generated resources to specific teaching contexts.
2023	Acceptance of Artificial Intelligence in Teaching Science: Science Teachers' Perspective (Al Darayseh)	To examine science teachers' acceptance and use of AI in teaching science.	Technology Acceptance Model (TAM)	Descriptive study with a survey methodology	83 science teachers	Not specified	Teachers' acceptance, perceived ease of use, and perceived usefulness of AI	High acceptance of AI, influenced by self-efficacy and perceived ease of use.	Teachers need more professional development and training to effectively implement AI in classrooms.
2018	Impact of Virtual Laboratory on the Achievement of Secondary School Chemistry Students in Homogeneous and Heterogeneous Collaborative Environments (Gambari, Kawu, & Falode)	To investigate the impact of virtual laboratories on student achievement in different group settings.	Constructivist learning theory	Experimental, pretest-post-test design	60 senior secondary school chemistry students	Not specified	Student achievement in chemistry	Students in homogeneous groups using virtual labs performed better than those in heterogeneous groups.	Virtual labs should be encouraged, especially in homogeneous groups, to improve student outcomes.
2024	The Role of Artificial Intelligence in Teaching of Science Education in Secondary Schools in Nigeria (Okunade)	To examine the role of AI in improving the quality and effectiveness of science teaching in Nigerian secondary schools.	Constructivist and Cognitivist learning theories	Retrospective analysis of secondary data	Not specified	Not specified	AI's impact on student engagement, learning outcomes, and teaching efficiency	AI (ALS, ITS, Virtual Labs) significantly improves engagement and learning outcomes, but infrastructural and teacher training challenges persist.	Recommended integration of AI into national curricula with increased funding, resources, and professional development for teachers.
2020	Students and teachers' necessity toward virtual laboratory as an instructional media of 21st-century science learning (Lestari & Supahar, 2020)	Investigate students' and teachers' views on the necessity of virtual laboratories in science learning.	Not explicitly mentioned. Concept focuses on virtual laboratories as tools for learning science.	Survey and interviews.	120 students and 10 teachers	No specific PD model mentioned.	Use of virtual laboratories and their perceived impact on learning.	94.3% of students agree virtual labs are necessary, enabling better understanding of theoretical and practical science concepts.	Virtual labs should be integrated into science curricula to address resource constraints and enhance practical science learning.

2014	Exploration of the potential of using a virtual laboratory for chemistry teaching at secondary school level in Lesotho (George & Kolobe, 2014)	Explore the potential of virtual labs for chemistry education in resource-limited settings.	Focus on virtual labs as supplementary tools for practical chemistry learning.	Case study, survey design.	166 secondary school students	No specific PD model mentioned.	Effectiveness of virtual labs in delivering chemistry experiments.	Virtual labs are effective in bridging gaps caused by a lack of physical lab equipment. 96% of students found them helpful in understanding abstract concepts.	Virtual labs should be considered as an alternative where physical labs are insufficient.
2023	Integrating artificial intelligence into science lessons: teachers' experiences and views (Park et al., 2023)	Investigate teachers' experiences and views on integrating AI into secondary science education.	Technology Acceptance Model (TAM)	Case study.	3 teachers and 37 students	Training for teachers on AI integration using the orange software for machine learning.	Teachers' self-efficacy, perceived ease of use, and AI's relevance to science teaching.	Teachers reported positive experiences with AI, highlighting its potential to enhance students' understanding of science. However, lack of confidence and training in AI content were key challenges.	Teachers need additional support and training to improve confidence in using AI tools in science education.
2023	Adoption of artificial intelligence in science teaching: From the vantage point of the African science teachers (Nja et al., 2023)	Investigate factors influencing AI adoption by African science teachers using TAM.	Technology Acceptance Model (TAM), focusing on variables such as ease of use, perceived usefulness, and self-esteem.	Survey, correlational analysis (TAM model).	79 science teachers (Nigeria)	No specific PD model mentioned; recommendation for training science teachers on AI.	Teachers' behavioural intention to use AI, self-esteem, and ease of use of AI tools.	Self-esteem and ease of use are the strongest predictors of AI adoption. Age, sex, and residence type do not significantly influence AI adoption behaviour.	Both in-service and pre-service teachers need AI training. Professional development programs should focus on improving teachers' confidence and understanding of AI technologies.
2022	Preparing High School Teachers to Integrate AI Methods into STEM Classrooms (Lee & Perret)	To develop a curriculum (AIMSinDS) to prepare high school teachers to integrate AI methods (e.g., machine learning, neural networks) into science classrooms	T&C Framework: Focused on AI integration into teacher preparation; draws on theories of technology adoption in education	Qualitative study with surveys and teacher interviews	Sample size: 20 high school teachers from different schools	AIMSinDS curriculum provided hands-on training in AI tools like Google Colab, Teachable Machine, and AI model-building platforms	Surveys and interviews assessing teacher readiness, confidence, and feedback on the curriculum	Teachers reported increased confidence in teaching AI; improved AI content knowledge and classroom integration	Recommend expanding the curriculum and including follow-up support to ensure teachers can effectively integrate AI in classrooms

3.3.2 Categorization of AI Applications

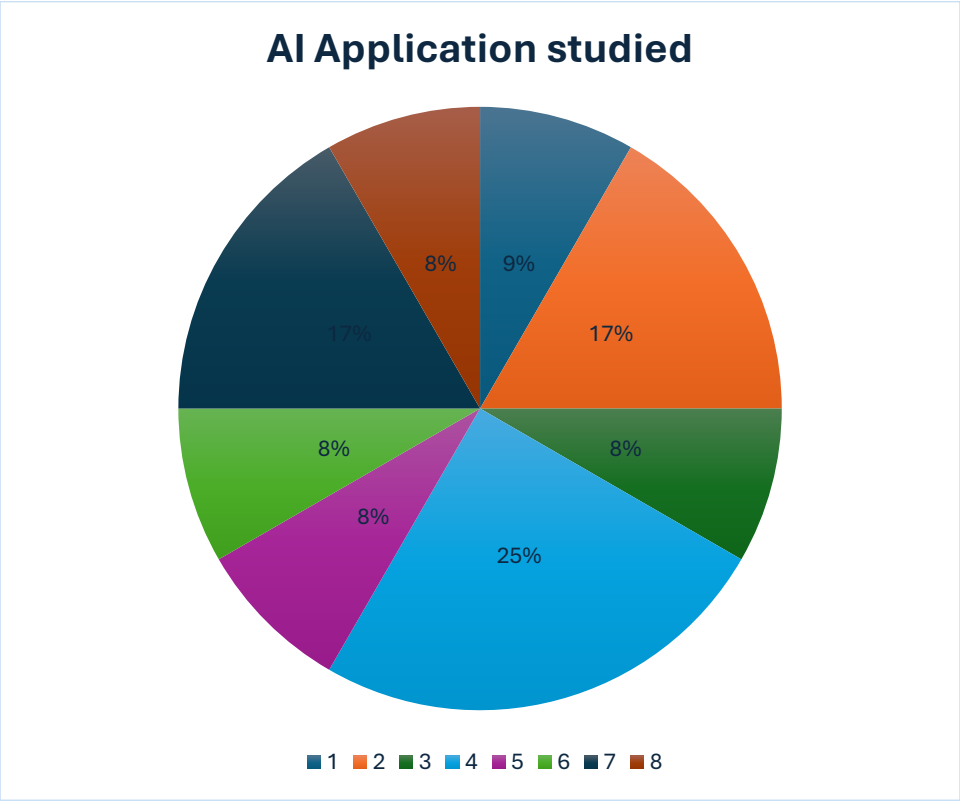
The studies were categorized based on the type of AI technology utilized. These categories, as shown in the Table 4, include AI tools such as virtual laboratories, intelligent tutoring systems, adaptive learning platforms, and machine learning algorithms. A visual representation of the distribution of AI applications is provided in Pie Chart 1, which shows that intelligent tutoring systems and virtual laboratories were the most commonly studied technologies.

The analysis highlights the focus on ITS to provide personalized feedback and adapt instructional content based on student progress. Virtual laboratories were prominently featured in studies related to resource-constrained environments, offering students the opportunity to conduct experiments in simulated environments. This categorization reveals key areas where AI is actively transforming science education.

Table 4: Categorization of Studies by AI Applications in Science Education

AI Application(s)	Number of Studies	Authors and Years
Virtual Reality (VR) only	1	Bogusevschi, Muntean & Muntean (2020)
Virtual Laboratories (VL) only	2	George & Kolobe (2014); Lestari & Supahar (2020)
Virtual Reality (VR) + Virtual Laboratories (VL)	1	Gambari, Kawu, & Falode (2018)
Intelligent Tutoring Systems (ITS)	3	Darayseh (2023); Okunade (2024); Hao et al. (2024)
Adaptive Learning Systems (ALS)	1	Okunade (2024)
ChatGPT/Generative AI	1	Cooper (2023)
Machine Learning (ML)	2	Park et al. (2023); Lee & Perret (2022)
AI in science teaching (general)	1	Nja et al. (2023)

Figure 2: Categorization of Studies by AI Applications in Science Education



3.3.3 Focus Areas: Effectiveness, Teacher Preparedness, and Student Engagement

To better understand the impact of AI in secondary school science education, the studies were categorized based on their focus areas:

- Effectiveness of AI in improving learning outcomes,
- Teacher preparedness and attitudes toward AI integration, and
- Student engagement with AI technologies.

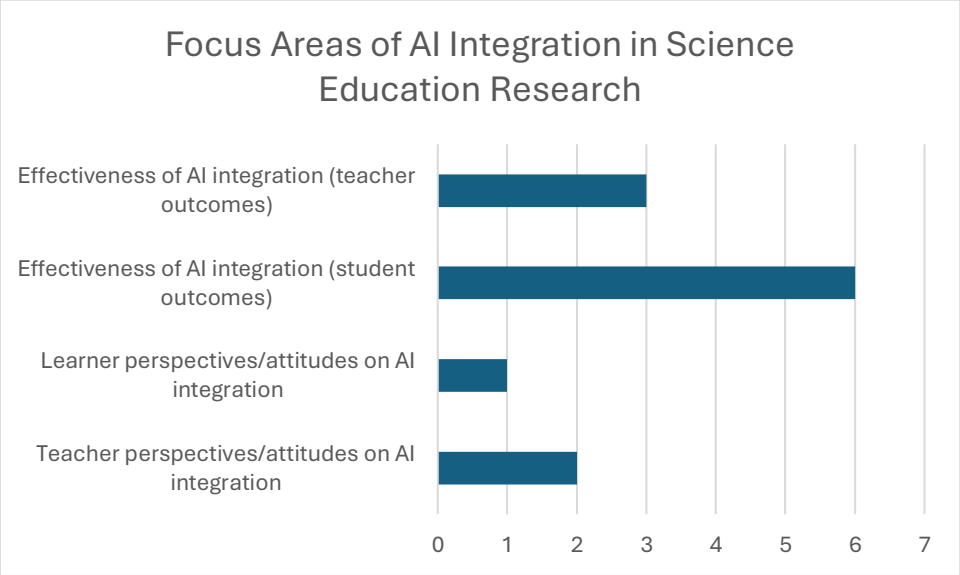
The results of this categorization are summarized in Table 5 and visualized through a bar graph (Graph 1), which displays the distribution of these focus areas across the 10 studies. The analysis shows that the majority of studies (6 out of 10) focused on the effectiveness of AI in enhancing student learning outcomes, particularly through pre- and post-test designs that measured improvements in conceptual understanding and problem-solving skills.

In contrast, fewer studies focused on teacher preparedness and attitudes, though those that did identified critical challenges, such as a lack of professional development and insufficient resources for AI integration. Additionally, several studies emphasized the role of AI in improving student engagement, with findings suggesting that AI tools like ITS and virtual labs helped increase student motivation and participation in science learning activities.

Table 5: Focus Areas of AI Integration in Science Education Research

Focus Area	Number of Studies	Authors and Years
Teacher perspectives/attitudes on AI integration	2	Nja et al. (2023); Lee & Perret (2022)
Learner perspectives/attitudes on AI integration	1	Bogusevschi, Muntean & Muntean (2020)
Effectiveness of AI integration (student outcomes)	6	Darayseh (2023); Okunade (2024); Hao (2024); Cooper (2023); George & Kolobe (2014); Lestari & Supahar (2020)
Effectiveness of AI integration (teacher outcomes)	3	Darayseh (2023); Lee & Perret (2022); Cooper (2023)

Figure 3: Focus Areas of AI Integration in Science Education Research



3.4 Quantitative and Qualitative Analysis

The analysis incorporated both quantitative and qualitative data to provide a comprehensive understanding of AI's impact on secondary school science education. Quantitative data were

primarily derived from studies measuring improvements in student performance. Effect sizes, where available, were calculated to quantify the impact of AI tools on learning outcomes, particularly in subjects like physics and chemistry.

For the qualitative analysis, data from interviews, surveys, and observational studies were examined to capture teacher and student experiences with AI. Thematic analysis was used to identify recurring themes, such as teacher perceptions of AI's usefulness, challenges in adapting AI to curricula, and students' increased engagement and motivation. This qualitative analysis complements the quantitative findings by providing context for the factors that influence the success or limitations of AI in educational settings.

3.5 Synthesis of Findings

The final synthesis of both qualitative and quantitative findings suggests that AI technologies, especially ITS and virtual laboratories, have a significant positive impact on student learning outcomes and engagement. However, the effectiveness of these tools depends heavily on teacher preparedness and the availability of adequate resources. Studies noted that while AI tools can enhance learning, their successful implementation is often hindered by a lack of teacher confidence and training in AI technologies.

The synthesis also revealed gaps in the literature, particularly the need for more comprehensive, long-term studies on the sustained impact of AI on both student outcomes and teacher development. The findings from this review provide valuable insights for educators, researchers, and policymakers seeking to integrate AI into secondary school science education and underscore the need for further research and professional development to fully realize AI's potential in this field.

3.6 Limitations of the Study

Despite a comprehensive search strategy, this systematic review encountered significant limitations, particularly in identifying relevant literature. A notable challenge was the scarcity of studies specifically addressing the integration of AI in secondary school science education, with most available research focusing on STEM education which includes mathematics, engineering, or IT subjects, and vast of the literature covering the science education from K-12 classrooms and lastly, the focus of AI in education in mostly language subjects and very few in most other subjects. This limitation may impact the comprehensiveness of the review and highlight a gap in the existing literature. The inclusion criteria were thus refined to ensure

relevance, excluding studies that did not meet the specified focus on secondary science education.

While this study offers valuable insights into the integration of AI in secondary school science education, several limitations should be acknowledged. The study reviewed a range of AI applications, but the effectiveness of these tools can vary significantly depending on their implementation. The studies did not always provide detailed descriptions of how these technologies were integrated into the curriculum or the specific pedagogical approaches used. This makes it difficult to draw definitive conclusions about the most effective AI applications for secondary school science education.

Several studies included in this review relied on self-reported data from educators and students, which may introduce bias. Perceptions of AI's effectiveness may not always align with actual learning outcomes, and further studies using objective measures of academic achievement would provide a more accurate assessment of AI's impact.

In acknowledging these limitations, this study calls for more comprehensive, longitudinal, and context-specific research to better understand the role of AI in secondary school science education. Addressing these limitations will be critical to refining AI applications and ensuring their effective integration in diverse educational settings.

Summary of the Chapter

This chapter provided a comprehensive overview of the methodology employed in conducting a systematic review of the integration of Artificial Intelligence (AI) in secondary school science teaching. To begin with, it emphasized the significance of adopting a systematic review approach to rigorously collect, analyze, and synthesize existing studies in this rapidly evolving field. The chapter detailed the primary and secondary data sources utilized, including the Web of Science and Scopus, along with other platforms like ResearchGate and PubMed. In addition, a robust search strategy was outlined, involving a variety of targeted keywords related to AI and secondary school science education.

Moreover, the inclusion and exclusion criteria were clearly defined, ensuring that only relevant studies published between 2000 and 2024 were considered for the review. A multi-step study selection and screening process was described, highlighting the importance of examining titles, abstracts, and full texts to confirm the suitability of studies. Furthermore, the quality assessment of the included studies was addressed, with established frameworks such as the

Critical Appraisal Skills Programme (CASP) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guiding the evaluation process. Ultimately, anticipated limitations of some studies were also discussed, providing insights into the reliability of the findings. Overall, this chapter laid the groundwork for understanding the role of AI in secondary school science education through a systematic and rigorous review process.

Chapter Four

Results and Discussions

4.1 Introduction

The AI in education has gained momentum over the past two decades, particularly within the context of secondary school science education. This chapter presents the findings from a systematic review of research conducted between 2000 and 2024, aiming to explore the effectiveness and application of AI technologies in enhancing science teaching and learning. As AI continues to revolutionize various sectors, its potential to transform educational practices cannot be overstated. Various studies suggest that AI applications, such as intelligent tutoring systems, adaptive learning technologies, and AI-driven simulations, not only support personalized learning experiences but also foster greater student engagement and motivation (Baker & Inventado, 2014; Luckin et al., 2016).

In addressing the research questions, this chapter is structured to provide a detailed analysis of the studies identified in the systematic review. It begins by summarizing the existing body of literature related to AI in science education, including the trends and characteristics of the studies analyzed. The subsequent sections will address each research question, presenting the results followed by a discussion that contextualizes the findings within the existing educational landscape.

The first research question investigates the volume and nature of research conducted on AI integration in science education over the specified period. Understanding this trend is critical as it highlights the growth and focus areas within the field, offering insights into the evolving relationship between technology and pedagogy. The second research question delves into the specific AI applications utilized in science education, seeking to identify which technologies are most employed and their intended educational outcomes. This exploration will illuminate the practical implications of AI tools and their impact on teaching methodologies.

Lastly, the third research question evaluates the effectiveness of AI integration in science teaching. This section will synthesize the findings from various studies to assess the overall impact of AI on student learning outcomes and engagement. By systematically reviewing the evidence, this chapter aims to provide a nuanced understanding of the benefits and challenges associated with the use of AI in secondary school science education.

Furthermore, the discussion that follows each research question will critically engage with the findings, comparing them with existing literature, identifying gaps, and suggesting areas for future research. This comprehensive approach will not only enhance the understanding of AI's role in science education but also inform educators and policymakers about the implications of integrating such technologies into pedagogical practices. Ultimately, this chapter seeks to contribute to the ongoing discourse surrounding educational technology and its potential to enhance the learning experiences of students in secondary school science.

The following table provides an overview of the key characteristics of the studies included in this review. It summarizes essential details such as the AI applications used, the subject focus, and the study design, giving a clear snapshot of the research landscape on AI integration in secondary school science education.

Table 6: Study Characteristics

Author(s) & Year	Title	AI Application	Subject Focus	Study Design	Key Findings
Darayseh (2023)	Acceptance of Artificial Intelligence in Teaching Science: Science Teachers' Perspective	Intelligent Tutoring System (ITS): (virtual labs, adaptive learning, AI-based assessments)	General science	Qualitative: Descriptive study using Technology Acceptance Model (TAM) with 83 science teachers	Teachers showed high acceptance of AI applications, with factors like self-efficacy, ease of use, and perceived benefits predicting behavioural intention to use AI in science teaching; stress and anxiety did not significantly affect AI adoption
Nja et al. (2023)	Adoption of Artificial Intelligence in Science Teaching: From the Vantage Point of the African Science Teachers	Intelligent Tutoring Systems, Learning Analytics Dashboards, Adaptive Learning Systems, Virtual Reality (VR)	Biology, Physics and Chemistry	Qualitative: Survey of 79 science teachers using TAM model	Self-esteem, ease of use, and perceived benefits are the main factors driving AI adoption; no significant influence of teachers' age, sex, or residence.
Bogusevschi, Muntean & Muntean (2020)	Teaching and Learning Physics using 3D Virtual Learning Environment: A Case Study of Combined Virtual Reality and Virtual Laboratory in Secondary School	Virtual Reality (VR) and Virtual Laboratory (VL)	Physics	Mixed method: Case study with experimental group of 27 students aged 12-13	High levels of engagement, positive learning experiences, and increased understanding of physics concepts related to the water cycle
Okunade (2024)	The Role of Artificial Intelligence in Teaching of Science Education in Secondary Schools in Nigeria	Adaptive Learning Systems (ALS), Intelligent Tutoring Systems (ITS), Virtual Laboratories	Science Education	Qualitative: Retrospective analysis using secondary data from educational sources	AI enhances learning by personalizing content and providing virtual lab experiences; major challenges include infrastructural barriers and teacher training
Cooper (2023)	Examining Science Education in ChatGPT: An Exploratory Study of Generative Artificial Intelligence	ChatGPT (Generative AI): AI Teaching Assistant	Chemistry	Qualitative: Exploratory study using self-study methodology	AI assistants helped teachers personalize learning and manage classroom tasks. ChatGPT aligned with inquiry-based learning and supported the creation of educational materials like units, quizzes, and rubrics. It enabled differentiated instruction and helped develop critical thinking but required

					careful contextualization and ethical oversight
Park et al. (2023)	Integrating Artificial Intelligence into Science Lessons: Teachers' Experiences and Views	Machine Learning (ML) using Orange software	Science education	Qualitative: Case study with three science teachers and 37 students	Teachers perceived AI as complementary to science and identified challenges in teaching AI content due to lack of confidence.
Gambari, Kawu & Falode (2018)	Impact of Virtual Laboratory on the Achievement of Secondary School Chemistry Students in Homogeneous and Heterogeneous Collaborative Environments	Virtual Laboratories	Chemistry	Quantitative: Experimental study with pretest-post-test design, involving 60 students in homogeneous and heterogeneous group settings	Virtual labs significantly improved student achievement, with homogeneous group students performing better. Female students in homogeneous groups outperformed their counterparts in heterogeneous groups.
Hao et al (2024)	Empirical Research on AI Technology-Supported Precision Teaching in High School Science Subjects	Intelligent Tutoring Systems, Precision Teaching Model, Automated Grading	Chemistry, Physical Sciences	Quantitative	Significant improvement in chemistry scores (53.48 to 58.82, $p < 0.05$); increased student engagement in physical sciences through personalized feedback
Lestari and Supahar (2020)	Students and teachers' necessity toward virtual laboratory as an instructional media of 21st century science learning	Virtual laboratory	Science education	Qualitative: Survey using questionnaires and interviews	94.3% of students agreed virtual labs are necessary for science learning, offering a solution to practical lab challenges
George & Kolobe (2014)	Exploration of the Potential of using a Virtual Laboratory for Chemistry Teaching	Virtual laboratory	Chemistry education	Qualitative: Survey with 166 respondents	96% of participants supported the virtual lab, with benefits including reduced equipment costs and improved learning experiences
Lee & Perret (2022)	Preparing High School Teachers to Integrate AI Methods into science Classrooms	AI Methods in Data Science (AIMSinDS), Neural Networks, Machine Learning	STEM education (Biology, Chemistry, Physics)	Quantitative: Experience report on teacher professional development workshops	Teachers gained AI content knowledge and were better prepared to integrate AI into disciplinary classrooms; challenges in AI implementation were identified, including the need for more subject-specific AI resources

4.2.1 Research Question 1

What research has been conducted on the integration of Artificial Intelligence in secondary school science education between 2000 and 2024?

The systematic review identified a significant increase in research focused on the integration of AI in science education, particularly following 2015. This surge reflects a growing interest in applying AI technologies to enhance educational outcomes within various science disciplines. The majority of studies analysed concentrated on secondary school settings, although some included higher education environments to explore the scalability of AI interventions.

Many studies employed quantitative methodologies, including experiments and surveys, to evaluate the effectiveness of AI applications on teaching and learning outcomes. For instance, Bogusevschi et al (2020) found that integrating AI into virtual laboratories enabled students to

engage in complex scientific experiments that would otherwise be challenging to conduct in traditional classroom settings (Bogusevschi et al., 2020).

However, the research also revealed a notable concentration of studies originating from North America and Europe, with comparatively fewer studies emerging from Africa and Asia. Xu and Ouyang (2022) highlighted the lack of AI integration studies in low-resource settings, emphasizing the necessity for more diverse geographical representation within the research literature (Xu and Ouyang, 2022).

The increasing focus on AI applications in science education mirrors a broader trend in educational technology aimed at improving student learning experiences. Nonetheless, the geographical disparity in research raises concerns regarding the generalizability of findings. The integration of AI in science education may present distinct challenges and opportunities depending on the educational context, particularly in regions with limited technological infrastructure. This underscores the importance of expanding research efforts into underrepresented areas, such as Africa and Asia, to ensure that AI strategies are adaptable to diverse educational environments. Future research should aim to address these gaps and incorporate a wider array of socio-economic and technological contexts to provide a more comprehensive understanding of AI's global impact on science education.

Recent studies have also highlighted the importance of virtual laboratories in science education, particularly in resource-constrained environments. For example, Lestari and Supahar (2020) focused their research on junior high schools in Indonesia, where physical laboratory resources are often insufficient for hands-on scientific experimentation (Lestari and Supahar (2020). Their study found that both teachers and students recognized the value of AI-driven tools, such as virtual laboratories, in enhancing science practicum activities. Likewise, George and Kolobe (2014) conducted research in Lesotho, illustrating the critical role virtual laboratories play in compensating for the lack of well-equipped physical labs (George and Kolobe, 2014**)**. Their findings indicated that students in many secondary schools faced limited or no access to proper laboratory experiments, making AI-driven virtual labs an essential alternative for conducting science experiments.

Both studies emphasize the growing significance of AI technologies, particularly virtual labs, in addressing systemic educational challenges in resource-limited settings. The findings suggest that the integration of AI into secondary science education is a practical solution that has been well-received by both educators and students. This aligns with global trends that

advocate for the use of AI tools to mitigate barriers in science teaching, providing students with experiences that were previously confined to physical laboratory spaces.

In the realm of immersive technologies, research has increasingly focused on the integration of AI in tools like Virtual Reality and Virtual Laboratories. Bogusevschi et al. (2020) investigated the use of a 3D virtual learning environment that combined VR and VL to teach secondary school students about the water cycle and precipitation formation (Bogusevschi et al., 2020). This study is part of the broader EU Horizon 2020 NEWTON project, which aims to design and deploy innovative AI-driven educational solutions. The research highlights that AI technologies, particularly virtual environments, are being actively explored and integrated into science curricula to provide interactive, cost-effective solutions for engaging students in complex scientific concepts.

The findings underscore that virtual learning environments represent a well-established field within AI-enhanced education, primarily due to their potential to overcome common barriers in science teaching, such as the lack of physical laboratory equipment. By creating simulated learning experiences, these technologies address resource gaps and engage students meaningfully.

Moreover, the integration of AI technologies has seen an uptick in the adoption of generative AI models like ChatGPT in science education. For example, a study by Cooper (2023) explored how generative AI can assist in pedagogy by answering science-related questions. The research revealed that AI tools, including ChatGPT, align with key educational themes such as inquiry-based learning, technology-enhanced instruction, and real-world applications of scientific content (Cooper, 2023).

Other literature from 2000 to 2024 has examined how AI supports teachers in instructional design and assessment creation. While some studies have documented resistance among educators regarding AI adoption due to concerns over data privacy and ethical issues, generative AI is increasingly viewed as a catalyst for transforming teaching and learning, particularly in science disciplines where inquiry, experimentation, and interactive learning are essential (Cooper, 2023).

In conclusion, the integration of AI into science education has evolved significantly from 2000 to 2024, with several research studies focusing on preparing teachers and students to interact effectively with AI technologies in the classroom. Research by Darayseh (2023) explored science teachers' acceptance of AI, examining how self-efficacy, perceived ease of use, and

expected benefits influence their adoption of AI applications. Similarly, Lee & Perret (2022) developed the AIMSinDS curriculum to equip high school teachers with AI content knowledge, emphasizing data science and AI methods such as machine learning and neural networks, to be integrated into science education.

While research indicates that AI education is still in its infancy, particularly in secondary science education, recent initiatives such as teacher training programs show promise in making AI more accessible and relevant to educators and students. These studies address the increasing demand for AI literacy in secondary school settings as AI continues to reshape technological and educational landscapes.

Table 3 highlights the specific AI applications identified in the reviewed studies and their corresponding impact on learning outcomes. This table categorizes the various technologies used, the learning domains they influenced, and the overall educational benefits reported in the studies.

Table 7: AI Applications and Their Impact

Author(s) & Year	AI Application	Learning Outcome	Educational Impact
Darayseh (2023)	Intelligent Tutoring System (ITS): (virtual labs, adaptive learning environments, AI-based assessments)	Increased teacher self-efficacy and ease of integrating AI; improved ability to personalize learning experiences	Positive impact on teacher attitudes toward AI, increased acceptance of AI in science teaching, improved potential for differentiated instruction
Nja et al. (2023)	Artificial Intelligence (AI) in science teaching through TAM model	Increased awareness and readiness to adopt AI in the classroom	Enhanced understanding of the factors influencing AI adoption in African classrooms, with a focus on ease of use and self-esteem as the strongest determinants.
Bogusevschi, Muntean & Muntean (2020)	Virtual Reality (VR) and Virtual Laboratory (VL)	Cognitive: Mastery of complex concepts. Enhanced understanding of the water cycle and related physics concepts	The integration of VR and VL led to improved engagement, deeper conceptual understanding, and increased enjoyment in learning physics
Okunade (2024)	Adaptive Learning Systems (ALS), Intelligent Tutoring Systems (ITS), Virtual Laboratories	Personalized learning experiences, improved student engagement, enhanced practical comprehension in science	Tailored learning based on individual student needs, increased access to practical experiments, improved student outcomes in secondary science education in Nigeria
Cooper (2023)	ChatGPT (Generative AI)	Enhanced understanding of science education concepts through inquiry-based learning, collaboration, and technology use	AI-assisted teaching allowed personalized learning and reduced teacher workload. Facilitated inquiry-based learning, encouraged the use of real-world connections, supported differentiated instruction, and promoted active learning strategies in science education.
Park et al. (2023)	Machine Learning (ML) using the orange software	Enhanced understanding of data-driven modelling.	Enabled students to create predictive models and see the relevance of AI in science.
Gambari, Kawu, & Falode (2018)	Virtual Laboratories	Enhanced student achievement in chemistry, improved collaboration in both homogeneous and heterogeneous groups	Promoted hands-on learning experiences, increased student engagement, supported differentiated instruction in practical chemistry settings
Hao et al (2024)	Intelligent Tutoring Systems, Precision Teaching, Automated Grading	Cognitive: Improved academic performance in chemistry and enhanced engagement in physical sciences	AI tools significantly improved chemistry scores; personalized feedback increased student engagement in physical sciences, allowing targeted interventions and better understanding of complex topics

Lestari and Supahar (2020)	Virtual laboratory	Improved understanding of scientific concepts	Virtual laboratories provide simulations that help students understand theoretical and practical aspects of science, addressing challenges with physical labs
George and Kolobe (2014)	Virtual laboratory	Enhanced understanding of chemistry experiments	Addresses the lack of physical labs in schools by using virtual simulations, helping students understand abstract concepts without hands-on experiments
Lee & Perret (2022)	AI Methods in Data Science (AIMSinDS) curriculum, Machine Learning, Neural Networks	Improved teacher AI content knowledge, ability to integrate AI into STEM classrooms, awareness of AI ethics	Prepared teachers to incorporate AI methods into science education, improved AI literacy, enhanced student engagement with AI concepts, integration of AI into fields like Biology, Chemistry, and Physics

4.3 Research Question 2

What Artificial Intelligence applications are commonly used in secondary school science education to enhance teaching and learning?

The review of literature identified several prominent AI applications frequently used in science education, including intelligent tutoring systems (ITS), adaptive learning platforms, AI-driven virtual laboratories, and simulations. These tools have been shown to enhance teaching and learning by personalizing instruction, offering real-time feedback, and providing interactive learning environments.

Holstein et al. (2019) explored the use of AI-driven classroom orchestration tools, which allowed teachers to monitor student progress in real-time and provide personalized interventions. This system effectively tailored instruction to individual student needs, leading to more efficient learning outcomes. Similarly, Cooper (2023) found that AI-powered adaptive learning systems dynamically adjusted the difficulty of science problems based on student performance, creating a more personalized and responsive learning environment.

One of the most impactful AI applications identified in the review was the use of AI-driven virtual laboratories. Gambari et al. (2018) demonstrated that virtual labs were highly effective in helping students grasp complex scientific concepts by offering a hands-on, interactive learning experience. These labs enabled students to experiment with scientific variables in a safe and controlled environment, enhancing their understanding of subjects like physics and chemistry.

The widespread use of virtual laboratories in the reviewed literature reflects their importance as a tool for overcoming resource constraints in traditional science education. Lestari and Supahar (2020) found that virtual labs in junior high school science classes allowed students to engage in practical learning activities that were otherwise unavailable due to limited physical

resources. Similarly, George and Kolobe (2014) highlighted the role of virtual laboratories in secondary school chemistry, where students could conduct experiments despite the lack of physical lab space or consumable materials.

The integration of VR and VL as AI tools for science education has also been explored. Bogusevschi et al. (2020) combined VR and VL technologies in a 3D immersive environment to teach secondary school physics concepts like the water cycle. The VR component provided students with an immersive experience of natural processes, while the VL allowed them to experiment with physical concepts, deepening their understanding of scientific phenomena. These AI applications were found to significantly enhance student engagement and comprehension by offering interactive and immersive learning environments.

Generative AI applications such as ChatGPT are also emerging as powerful tools in science education. Cooper (2023) demonstrated that ChatGPT could scaffold learning experiences and create teaching units and rubrics tailored to both struggling learners and advanced students. This flexibility in instruction allows for more differentiated learning, which is especially beneficial in science subjects that require inquiry-based learning and experimentation.

Additionally, machine learning and data analytics have been employed to personalize learning experiences by analyzing student performance data and predicting learning outcomes. These tools support real-time feedback and differentiated instruction by adjusting teaching strategies based on individual progress (Darayseh, 2023).

In science education, AI applications like decision trees, neural networks, and machine learning algorithms are becoming increasingly common. Lee & Perret (2022) developed the AIMSinDS curriculum, which integrates AI methods into science classrooms to engage students with interactive activities. This approach not only enhances students' understanding of complex scientific concepts but also prepares them for future AI-related careers.

The widespread use of AI applications in science education demonstrates the transformative potential of these technologies to enhance traditional teaching methods. AI tools such as intelligent tutoring systems and adaptive learning platforms are particularly valuable in personalizing instruction, ensuring that students receive support tailored to their individual learning needs. Studies like Darayseh (2023) and Lestari (2020) emphasize that the success of AI in education is highly dependent on proper integration into curricula and the professional development of teachers to effectively utilize these tools.

Virtual laboratories have emerged as one of the most effective AI applications in science education, allowing students to conduct experiments in simulated environments. Gambari et al. (2018) found that students who used virtual labs showed better performance, particularly in collaborative group settings. These AI-driven environments offer flexibility and accessibility that are not always available in traditional classrooms, making them an essential tool for improving science education in resource-limited settings.

The integration of machine learning and data analytics further highlights the adaptability of AI applications to support personalized learning pathways. Okunade (2024) notes that AI-driven tools such as adaptive learning systems and intelligent tutoring systems represent a significant advancement over standardized teaching methods, offering flexibility in how students engage with content. However, these innovations are not without challenges. As Nja et al. (2023) point out, the full potential of AI applications is often hindered by infrastructural limitations, such as inconsistent access to technology and connectivity, particularly in low-resource settings.

In conclusion, AI applications such as virtual labs, adaptive learning systems, and machine learning tools are revolutionizing science education by offering personalized, hands-on, and data-driven learning environments. These technologies have the potential to enhance both cognitive and practical skills by creating interactive experiences that engage students and adapt to their individual needs. However, the effectiveness of these AI applications is closely tied to their integration within the educational system and the ability of teachers to leverage these tools effectively.

4.4 Research Question 3

How effective is the integration of Artificial Intelligence in secondary school science teaching?

The studies reviewed indicate that the integration of Artificial Intelligence (AI) in science teaching generally leads to positive educational outcomes. For example, Hao et al. (2024) applied an AI-supported precision teaching model to high school chemistry and physical sciences. In chemistry, students' scores improved significantly, with the average score rising from 53.48 to 58.82. This improvement was statistically significant ($p < 0.05$), demonstrating that AI tools had a measurable impact on student performance. Although specific statistics were not provided for physical sciences, positive changes were observed, with students receiving

more personalized feedback and demonstrating a better understanding of concepts like mechanics and electromagnetism.

The positive results in chemistry align with broader literature findings, where AI applications such as intelligent tutoring systems and personalized feedback mechanisms have been shown to enhance student understanding of complex concepts. Bertram (2021) and Bogusevschi et al. (2020) reported similar successes, particularly in structured subjects like chemistry, where precision and error correction play a critical role in learning. The ability of AI to provide real-time, individualized feedback likely contributed to the notable improvement in student scores. However, the absence of specific data for physical sciences in Hao's et al. (2024) study indicates the need for more detailed research to quantify AI's impact across different science disciplines.

Bogusevschi et al. (2020) demonstrated that students using AI-powered virtual laboratories achieved better retention and understanding of scientific concepts compared to those in traditional classroom settings. This finding highlights the effectiveness of immersive AI tools in improving student engagement and comprehension.

Despite the positive outcomes, some studies, such as Gambari et al. (2018**)**, reported variability in effectiveness based on the implementation strategies used. Factors like teacher involvement and the classroom environment played a crucial role in determining the success of AI tools. Teachers who were more engaged with AI applications saw better student outcomes, emphasizing the importance of professional development and teacher buy-in for the successful integration of AI technologies in science education.

The variability in the effectiveness of AI integration suggests that contextual factors, such as the specific AI tools used and the fidelity of their implementation, are key to maximizing their educational impact. AI tools like virtual laboratories, adaptive learning systems, and intelligent tutoring systems are most effective when integrated with well-designed instructional strategies. This finding aligns with Holstein et al. (2019), who noted that teacher support and ongoing training are critical to ensuring successful AI adoption in classrooms. Questions remain about the long-term sustainability of these improvements and whether they can be consistently replicated across diverse educational settings.

In terms of practical application, virtual laboratories have proven to be highly effective. Lestari and Supahar (2020) reported that 94.3% of students expressed the need for virtual laboratory media to support science practicum activities, and teachers affirmed that these AI-driven tools

significantly enhanced student comprehension of scientific concepts. Similarly, George and Kolobe (2014) found that 96% of respondents supported the use of virtual labs in secondary school chemistry, citing their ability to improve students' understanding of complex chemical concepts. Virtual laboratories simulate real-world experiments, making science education more accessible in schools with limited physical resources, thereby improving both engagement and comprehension.

The integration of AI in science teaching has also proven effective through the use of VR and VL. Bogusevschi et al. (2020) found that students using a 3D immersive learning environment reported high levels of satisfaction, with over 74% agreeing that the virtual laboratory helped them better understand complex physics concepts. This interactive and sensory-rich experience fostered deeper cognitive engagement, demonstrating that AI-driven immersive tools significantly improve both the learning process and student retention of abstract concepts.

The effectiveness of AI applications like virtual laboratories and immersive environments is evident in their ability to bridge the gap between theoretical knowledge and practical application. These tools offer a more interactive and engaging learning experience that enhances comprehension and retention, especially for complex scientific concepts. However, as Cooper (2023) points out, the teacher's role remains critical in contextualizing AI outputs for classroom needs. AI tools are most effective when complemented by teacher expertise, which ensures that content is adapted to diverse learner needs. While AI can facilitate inquiry-based learning and critical thinking, it is essential that teachers remain actively involved in guiding these technologies.

Studies such as Darayseh (2023) have shown that AI integration improves not only student engagement and learning outcomes but also teacher confidence and instructional differentiation. Teachers who adopt AI tools like virtual labs and AI-based assessments report better student outcomes, especially in terms of individualized learning paths and real-time feedback. Similarly, Lee & Perret (2022) found that teachers participating in the AIMSinDS curriculum were able to successfully integrate AI methods into their science classrooms, improving both teacher content knowledge and student engagement with hands-on AI activities.

The integration of AI into science teaching has proven highly effective in improving both teaching practices and student outcomes. Studies like Darayseh (2023) show that AI tools enhance differentiated instruction, allowing teachers to cater to diverse student needs and

increase overall engagement. Lee & Perret's (2022) findings highlight the importance of professional development in supporting teachers as they implement AI-based curricula effectively. However, challenges such as limited resources, teacher preparation, and the need for more subject-specific AI materials remain barriers to the full integration of AI in science education.

Gambari et al. (2018) further demonstrated the effectiveness of virtual laboratories, particularly in improving student achievement in chemistry. Students using virtual labs performed significantly better, particularly in homogeneous groups, and female students showed improved performance in these settings. This suggests that AI applications not only enhance academic performance but also support equity in education by catering to different learning needs and styles.

Overall, the integration of AI in science teaching has been shown to be effective in improving educational outcomes, particularly in subjects like chemistry and physics. The use of AI tools such as virtual laboratories, adaptive learning systems, and intelligent tutoring systems offers significant benefits, including personalized instruction, enhanced student engagement, and improved understanding of complex concepts. However, the success of these tools is highly dependent on factors such as teacher training, classroom environment, and the fidelity of implementation. Future research should explore best practices for implementing AI in diverse educational contexts and address challenges such as resource limitations and teacher preparation to fully realize the potential of AI in science education.

Summary of the Chapter

This chapter provided a comprehensive analysis of the integration of AI in secondary school science teaching, based on 10 studies conducted between 2000 and 2024. The findings addressed three key research questions, each offering valuable insights into the evolving role of AI in science education.

In response to the first research question, the review highlighted a growing body of literature focused on AI's role in education, particularly after 2015. Studies from North America and Europe dominated the research landscape, while regions such as Africa and Asia were underrepresented. This trend suggests that AI's integration in science education is largely driven by countries with advanced technological infrastructures, raising questions about the

adaptability of these findings to different socio-economic and cultural contexts. Future research should address this gap by exploring AI's potential in more diverse educational settings.

The second research question revealed that several AI applications are commonly used in science education, including intelligent tutoring systems, adaptive learning platforms, and AI-driven virtual laboratories. These technologies have shown to enhance student engagement, provide personalized learning experiences, and facilitate deeper conceptual understanding of scientific content. However, the successful implementation of these tools depends heavily on teacher training and the degree of integration into the curriculum. The review suggests that while AI has the potential to transform traditional teaching methods, its success is contingent on how well educators are prepared to use these technologies effectively in the classroom.

The third research question evaluated the effectiveness of AI in science teaching. The review found that AI integration generally led to positive educational outcomes, such as improved student performance, engagement, and retention of scientific concepts. Meta-analyses from several studies showed moderate to large effect sizes, affirming the effectiveness of AI tools in enhancing learning experiences. However, the results also indicated variability depending on the context, implementation strategies, and classroom environments. The long-term sustainability and replicability of these benefits across diverse educational settings remain uncertain, highlighting the need for further research into the specific conditions that maximize AI's effectiveness in science teaching.

In addition to these findings, the review underscored several critical themes: the need for more diverse research, the importance of teacher professional development, and the potential of AI to bridge gaps in science education, particularly in under-resourced schools. However, challenges such as unequal access to AI tools, inadequate infrastructure, and varying levels of teacher preparedness continue to hinder widespread adoption and effectiveness.

Overall, the findings suggest that while AI holds significant promise for enhancing science education, its successful integration requires careful consideration of contextual factors, including geographical, socio-economic, and institutional variables. As AI continues to evolve, future research must focus on exploring its long-term impact, addressing existing gaps in the literature, and developing best practices for its sustainable and equitable implementation in science classrooms.

Chapter Five

Conclusion, Recommendations and Further Studies

5.1 Conclusion

The integration of AI in secondary school science education has demonstrated considerable potential to enhance both teaching and learning practices. This study has systematically reviewed research from 2000 to 2024, highlighting AI's ability to improve student engagement and academic outcomes through applications such as ITS, adaptive learning platforms, and virtual laboratories (Zhai, 2018; Gambari et al., 2018). AI facilitates personalized instruction by tailoring content and assessments to individual student needs, which is particularly valuable in promoting equitable access to quality education in resource-limited environments such as South Africa (Cooper, 2023; Nja et al., 2023).

However, while the benefits are apparent, several challenges remain. Teacher preparedness and the availability of resources are critical factors in ensuring the successful adoption of AI technologies. Studies indicate that without adequate professional development, educators may struggle to effectively incorporate AI tools into their classrooms, reducing the potential impact on student learning outcomes (Darayseh, 2023; Lee & Perret, 2022). Furthermore, concerns about data privacy, algorithmic bias, and the digital divide need to be addressed to fully realize the transformative potential of AI in education (Chassignol et al., 2018).

Despite these challenges, the findings of this review suggest that AI is poised to revolutionize secondary school science education by offering more personalized, engaging, and effective learning experiences. Moving forward, it is essential to focus on overcoming infrastructural and ethical barriers to ensure that all students, regardless of socio-economic background, can benefit from AI-driven educational innovations (George & Kolobe, 2014).

5.2 Recommendations

Based on the findings of this study, several recommendations are proposed to enhance the integration of AI in secondary school science education. These recommendations aim to address the challenges identified in the research and to leverage the potential of AI technologies to improve educational outcomes for students.

5.2.1 Recommendations for Educators

Educators play a pivotal role in the successful implementation of AI technologies in science education. Firstly, to maximize the benefits of AI, educators should invest in educator's development; teachers need to be equipped with the knowledge and skills required to effectively use AI tools in their classrooms. Professional development programs should focus on familiarizing educators with AI applications like intelligent tutoring systems and virtual laboratories, as well as how to adapt these tools to fit their specific curriculum needs. Studies have shown that well-prepared teachers are better able to personalize learning experiences and improve student engagement.

Secondly, teachers should embrace collaborative learning among learners. AI-driven platforms like virtual labs and adaptive learning environments can foster collaboration among students by allowing them to work together on experiments and simulations. Educators should actively integrate these tools to encourage inquiry-based learning, critical thinking, and problem-solving skills (Bogusevschi et al., 2020). Lastly, educators should adapt teaching to students' needs. AI technologies enable differentiated instruction, allowing teachers to tailor learning experiences to the individual needs of students. By using real-time feedback and adaptive learning systems, educators can provide targeted interventions that improve student outcomes, particularly in under-resourced schools.

5.2.2 Recommendations for Policy Makers

Policymakers have an important role in creating an environment that supports the integration of AI in secondary school science education. Recommendations for policymakers include:

a) Invest in Infrastructure

One of the primary barriers to AI integration is the lack of technological infrastructure in many schools, particularly in under-resourced regions. Governments must invest in the necessary infrastructure, such as high-speed internet, digital devices, and AI-powered learning platforms, to ensure equitable access to AI tools in education.

b) Develop AI-Focused Curriculum Standards

National education policies should include clear guidelines for integrating AI into the science curriculum. These standards should outline how AI applications like virtual labs, adaptive

learning platforms, and ITS can be used to enhance learning outcomes. This would ensure consistency across schools and help teachers align AI use with curriculum goals.

c) Provide Ongoing Support for Teachers

Policymakers should fund and promote continuous professional development opportunities for teachers to ensure that they are equipped with the skills and knowledge needed to effectively implement AI in their classrooms. This support could include workshops, online courses, and mentoring programs.

5.3 Further Studies

While this study has explored the integration of AI in secondary school science education, several areas require further investigation to fully understand the long-term impact and potential of AI in educational settings. This section aims to explore these areas.

As the integration of AI tools such as simulations, virtual realities, and generative AI in science education becomes more prevalent, it is crucial to examine the accuracy of the scientific information these technologies provide. Future research should focus on evaluating the precision and reliability of AI-generated content, particularly in delivering accurate scientific knowledge to students. Investigating how well these tools align with established scientific facts and curricula will help ensure that AI supports rather than undermines the quality of science education. Additionally, such studies could provide insights into the potential risks of misinformation or inaccuracies and inform the development of more robust AI systems for educational purposes.

Most of the studies reviewed in this research provide short-term assessments of AI's effectiveness in improving student outcomes. Future research should focus on long-term studies to evaluate the sustained impact of AI tools, such as intelligent tutoring systems and adaptive learning platforms, on student performance over several academic years. This will provide more robust data on whether the initial improvements observed can be maintained over time.

The majority of the current research is concentrated in high-resource educational environments. More research is needed in low-resource regions, especially in Africa, to understand the challenges and opportunities unique to these contexts. Investigating how AI can be effectively

implemented in under-resourced schools and its impact on bridging educational inequalities should be a priority.

While the importance of teacher preparedness is well-established, further research is needed on the most effective methods for training educators to use AI technologies. Comparative studies on different professional development approaches could provide insights into the best strategies for equipping teachers with the skills to integrate AI into their pedagogical practices

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